

Climate resilience through bioeconomy: A mixed-methods protocol for assessing adaptation policies in rural settlements on the amazon

--Manuscript Draft--

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Abstract:	Introduction Climate change intensifies social, economic, and environmental vulnerabilities in Amazonian rural communities, where dependence on natural resources and limited institutional capacity constrain adaptation. The bioeconomy is often proposed as an alternative pathway for sustainable development, yet empirical evidence of its role in resilience remains scarce. Objective This study protocol presents a mixed methods design to evaluate climate adaptation policies and community resilience strategies in agrarian reform settlements, rural territories created through Brazil's land redistribution program, in Pará, Brazil. Although centered on a regional case, the protocol is structured to generate insights applicable to other socioecologically vulnerable rural contexts in the Amazon and comparable regions worldwide. Methods The study will be conducted in 15 municipalities across the three regional superintendencies of the National Institute for Colonization and Agrarian Reform. The design integrates (i) secondary data on socioeconomic, climatic, and policy indicators; (ii) semi-structured interviews with municipal managers (n = 1–3 per municipality); (iii) focus groups with community leaders (n = 8–12 per settlement); and (iv) household surveys (n = 20–50 families per settlement). Primary outcomes include the Income and Livelihood Diversification Index (ILDI), Climate Vulnerability Perception Scale (CVPS), Bioeconomy Engagement Index (BEI), and Reported Adaptive Capacity Index (RACI). Secondary measures include the Adaptation Strategy Diversity Index (ASDI) and a Municipal Climate Engagement Typology (MCET). Quantitative analyses will employ multilevel linear and generalized linear models (LMM/GLMM), while qualitative data will be analyzed through reflexive thematic analysis, with final integration achieved via a triangulation framework. Expected results This study will generate a comprehensive framework integrating indicators of vulnerability, adaptive capacity, and bioeconomy engagement across agrarian reform settlements in the Eastern Amazon. By combining household surveys, qualitative fieldwork, and municipal-level policy analysis, the protocol provides a transparent, replicable approach for assessing local climate adaptation processes. The study is designed to inform municipal and state policymakers, support the strengthening of local adaptation initiatives, and contribute evidence relevant to Sustainable Development Goals (SDGs 2, 13, and 15).
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	Vitor Castro
	Emilio Mendes
Response to Reviewers:	Dear Dr. Parker, Thank you for your message and for the opportunity to revise our manuscript. We appreciate the thorough guidance provided and have carefully addressed every editorial and reviewer requirement. Below we summarize the actions we have taken in response to the journal's technical instructions: Manuscript Formatting: We revised the manuscript to ensure full compliance with PLOS ONE's formatting and file-naming guidelines. The revised manuscript ("Manuscript"), the tracked-changes version, and the "Response to Reviewers" letter have been uploaded as separate files. Ethics Statement Placement: The ethics statement now appears only once, under the "Materials and Methods" section, as requested. All other mentions were removed. Data Availability Statement: We have updated the Data Availability Statement in the submission form. As this is a study protocol without primary data, we used the recommended formulation: "All relevant information is contained within the manuscript and its Supporting Information files. Future data involving human participants will be shared in fully anonymized, aggregate form in accordance with ethics approval and participant consent." Data Sharing Clarification: We adjusted the statement that previously indicated data would be made available upon acceptance, to reflect realistic ethical and legal constraints related to future human-participant data collection. Funding & Financial Disclosure Consistency: We corrected both the Funding Information and Financial Disclosure sections to contain identical information, using the following wording: "This study was funded by the National Council for Scientific and Technological Development (CNPq), Brazil, under Grant No. 445058/2024-2. The funder had no role in study design, data collection and analysis, decision to publish, or preparation of the manuscript." Removal of Identifiable Information from Supporting Files: As instructed, we removed the ZIP file previously uploaded as Supporting Information. The documents it contained (funding letter and ethics approval) have been re-uploaded individually under the category "For Editors Only / Not for Publication", since they contain author-identifying information required for editorial verification but are not meant for public dissemination. No participant data are included in these documents. Reviewer-Suggested Citations: We reviewed all suggested publications and incorporated those that were relevant and strengthened the manuscript. We hope that these revisions address all journal requirements. We are grateful for the constructive feedback and remain available for any further clarification. Thank you for your time and consideration. REVIEWER #1 We would like to thank you for the careful reading of our manuscript and for the constructive and insightful comments. We appreciate the positive evaluation regarding the clarity, relevance, and overall structure of the study. Following your guidance, we have revised the manuscript extensively. The changes have substantially improved methodological transparency, theoretical grounding, clarity of objectives, and the

international relevance of the protocol.
Below we provide a detailed, point-by-point response to all comments. All revisions have been incorporated in the updated manuscript.

1. Need for more citations in the Methods section
Reviewer comment:
“There is a lack of citations along the methods section... especially regarding the calculation of indexes and approaches for missing data analysis...”
Response:
We agree and have added multiple methodological references throughout the Methods section, including citations related to diversification measures, reflective and formative index construction, PCA, mixed-effects modeling, multiple imputation, and multilevel missing data handling. These additions strengthen transparency and replicability.

2. Need to summarize / merge certain sections (Data management, Ethics, Timeline)
Reviewer comment:
“Some sections in the methods could be significantly summarised and merged... data management, ethical considerations, project timeline.”
Response:
We have condensed these sections substantially:
•Ethical Considerations was moved to the Methods section and shortened.
•Data Management was reduced to a concise paragraph describing anonymization, storage, and OSF deposition.
•Study Status and Timeline was reduced and summarized in a brief paragraph.
This improves the manuscript's focus on scientific content.

3. Discussion needs expansion (design rationale, contributions, international relevance)
Reviewer comment:
“The discussion needs to be significantly expanded... explain why the chosen study design is appropriate... highlight how the protocol builds upon previous studies... describe international relevance...”
Response:
The Discussion has been substantially expanded to include:
•A clear justification for the mixed-methods, multilevel design.
•An explanation of alternative designs (e.g., panel, quasi-experimental) and why they were not feasible.
•A comparison with existing literature showing how our protocol integrates climate adaptation, bioeconomy, governance, and resilience.
•A new section on the international relevance of the study and pathways for generalization.
•A description of how limitations will be accessed and communicated in the final results.
Sections that were not directly scientific (e.g., dissemination plan, early termination) were removed.

4. Need for a subsection describing qualitative data collection
Reviewer comment:
“There should be a subsection prior to ‘Sample inclusion...’ that describes them.”
Response:
We added a new subsection titled “Data Collection Methods” before the sampling description, clarifying procedures for:
•household surveys
•semi-structured interviews
•focus groups
This ensures coherence with the qualitative saturation criteria.

5. Clarification of agrarian reform settlements and ‘settled families’
Reviewer comment:
The reviewer did not understand these terms.
Response:
We added a clear definition in the Introduction and Sample section explaining that agrarian reform settlements are rural territories created through Brazil’s federal land redistribution program (INCRA), and “settled families” refers to beneficiary households

receiving land through this program. This addresses ambiguity for international readers.

6. Clarifications on indicator descriptions (range, direction, origin, aggregation)

Reviewer comment:

The reviewer requested more details for ILDI, CVPS, BEI, RACI, MCET.

Response:

We revised all indicators to include:

- range of values
- expected direction (associational)
- conceptual grounding
- whether original or adapted
- aggregation from household to settlement level (when applicable)

Descriptions are now consistent and transparent.

7. Avoiding overstatements (e.g., “no standardized methodology”)

Reviewer comment:

The initial phrasing overstated the absence of methods.

Response:

We rewrote this section to present a more nuanced description, noting that existing methods are fragmented and unevenly applied, and clarifying that our protocol contributes by integrating climate adaptation, resilience, and bioeconomy in a multilevel framework.

Conclusion

We thank the reviewer and Academic Editor once again for the constructive feedback that greatly improved the clarity and contribution of the protocol. We respectfully resubmit the revised manuscript for further consideration.

REVIEWER #2

We thank you sincerely for your detailed and insightful review. Your comments significantly improved the methodological clarity, rigor, and overall contribution of our study protocol. We have carefully addressed every point raised and revised the manuscript accordingly. Below we provide a point-by-point response, following the structure of your comments.

1. Clarification of causal vs. associational language

Reviewer comment:

“The current framing occasionally suggests causal claims... Unless you add a credible identification strategy, the objectives and hypotheses should be clearly associational.”

Response:

We agree and have revised the entire manuscript to remove any causal language.

Specifically:

- All hypotheses (PH1–PH5) were rewritten using associational terms (“associated with,” “co-occur with,” “positively/negatively correlated with”).
- We removed verbs implying causality (“leads to,” “results in,” “increases,” “reflects”), replacing them with neutral correlational phrasing.
- The Objectives section was revised to explicitly state that the study aims to identify associations, not causal effects.
- We clarified the observational nature of the design in the Study Limitations.

These changes ensure full alignment with observational study and PLOS ONE guidance.

2. Pre-specification of confounders across levels & sensitivity analysis (Cinelli & Hazlett)

Reviewer comment:

“Pre-specify rich confounders... adopt Carlos Cinelli’s framework for sensitivity analysis... robustness values and contour diagnostics.”

Response:

We thank the reviewer for this excellent suggestion. We incorporated the following revisions:

- Added a new subsection explicitly listing confounders at the household, settlement, and municipal levels (market access, infrastructure, ecological context, cooperatives,

local associations, institutional capacity, political economy, and geographic factors).

- Added a detailed Sensitivity Analysis Plan, adopting the Cinelli & Hazlett 1 framework, with robustness values and contour plots to quantify the impact of unobserved confounding.
- Specified that analyses will be conducted using sensemakr, and results will report E-values/robustness thresholds.
- Clarified that this sensitivity analysis complements multilevel models and multiple imputation procedures.

These additions substantially strengthen the transparency and rigor of the analytical plan.

3. Expanded power calculations for multilevel design

Reviewer comment:

"Power deserves expansion to reflect clustering... ICC around 0.10... variation between communities... ordinal BEI harder to power... redo power and MDE calculations... grid of plausible ICCs... consider number of communities vs. number of respondents."

Response:

We fully revised the Power Calculations section to incorporate:

- A clear explanation of the role of ICCs and design effects.
- Simulations for continuous, binary, ordinal (CLMM), and count outcomes across ICCs = 0.05, 0.10, 0.20, 0.30.
- A statement that power is driven primarily by the number of settlements, not households.
- Justification for including 20–30 settlements in the sampling design.
- Minimum detectable effect estimates for:
 - oLMM (continuous),
 - oCLMM (BEI ordinal predictor/outcome),
 - oand logistic models (binary outcomes).
- Discussion of trade-offs (increasing clusters > increasing respondents).

We revised the introductory sentence to emphasize that power calculations were used to determine the minimum sample size required, not merely to illustrate tool usage.

4. Explicit clarification of where variation is expected to come from

Reviewer comment:

"It would help to be explicit about where you expect variation to come from... most identifying variation will be between communities."

Response:

We added explicit statements in both the Sampling Strategy and Quantitative Analysis sections that:

- Most identifying variation is expected at the settlement (level-2) scale, given the importance of market integration, cooperative presence, mobility, and ecological context.
- This expectation guided our choice to maximize the number of settlements and informed the multilevel modeling approach.

This cross-referencing improves internal coherence.

5. Clarification of ordinal modeling for BEI

Reviewer comment:

"The move from a binary to a four-category engagement index is sensible... commit to an analysis approach appropriate for ordered categorical."

Response:

We revised the Analytical Plan to state explicitly:

- BEI will be modeled using cumulative link mixed models (CLMM) when used as an outcome.
- BEI may serve as a level-1 or level-2 predictor, with appropriate aggregation (means or distributions) at the settlement level when required.
- The proportional odds assumption will be checked.

This ensures appropriate treatment of the indicator.

6. Missing data plan, with clarity across levels

Reviewer comment:

"Please indicate how you will handle missingness at each level... multiple imputation respecting clustering... reweighting strategies... sensitivity."

	Response: We expanded the Missing Data section by: •Differentiating missingness across household, settlement, and municipal levels. •Specifying multilevel multiple imputation using mice and pan, with clustering preserved. •Including inverse probability weighting (IPW) and augmented IPW for differential nonresponse. •Adding procedures for secondary data missingness (climate indicators, municipal plans, budgets). •Clarifying that missingness in municipal plans/budgets may be structural and analytically meaningful. •Integrating this with the Cinelli-style sensitivity analysis. This strengthens the rigor of the protocol. 7. Reduction of researcher degrees of freedom and primary outcomes Reviewer comment: “Pre-specifying a short list of primary outcomes and hypotheses... identify explicitly which variables will be used.” Response: We created a new paragraph at the beginning of Primary Outcomes: •Specifying exactly four primary outcomes (ILDI, CVPS, BEI, RACI). •Listing which hypotheses (PH1–PH5) each primary outcome relates to. •Clearly separating expected direction (conceptual meaning) from hypothesis testing (associational relationships). Secondary outcomes are now explicitly labeled and described as complementary. 8. Qualitative standards (COREQ/SRQR) and mixed-methods integration (GRAMMS) Reviewer comment: “Commit to COREQ/SRQR... provide a GRAMMS-style description... triangulation table.” Response: We added: •A clear statement that qualitative reporting will follow COREQ. •A full GRAMMS-consistent description of integration procedures. •A triangulation table with decision rules for convergence, complementarity, and discordance. •Updates to the Saturation and Qualitative Methods sections. These revisions significantly clarify the role of qualitative data. Conclusion We sincerely appreciate the reviewer’s contributions. The manuscript has improved substantially because of this detailed and thoughtful review. We believe the revised version now aligns fully with PLOS ONE’s expectations for study protocols and provides a stronger, clearer, and more rigorous analytical framework. We hope the revised manuscript meets with your approval, and we thank you again for your constructive feedback. Sincerely, Corresponding Author
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Competing Interests Use the instructions below to enter a competing interest statement for this submission. On behalf of all authors, disclose any competing interests that could be perceived to bias this work—acknowledging all financial support and any other relevant financial or non-financial competing interests.	The authors declare no competing interests. The study received public research funding from the National Council for Scientific and Technological Development (CNPq), Brazil (Grant No. 445058/2024-2), which had no role in study design, data collection and analysis, decision to publish, or preparation of the manuscript

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This study involves human participants and was reviewed and approved by the Research Ethics Committee (CEP) of the Federal University of Pará (UFPA), in accordance with Brazilian National Health Council Resolution No. 510/2016. The approval number is CAAE: 87492325.6.0000.0018.

Informed consent will be obtained from all participants prior to their inclusion in the study. For community members, the consent process will involve providing the Informed Consent Form (ICF) in writing, reading it aloud, and explaining it in clear and accessible language, considering possible variations in literacy levels; written consent will then be obtained by signature or fingerprint. For public managers, written consent will be obtained.

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This manuscript reports a study protocol and does not include the results of the pilot study. The pilot data were collected solely to refine instruments and procedures and cannot be publicly shared due to human participant privacy and ethical restrictions. Future study data will be shared in de-identified, aggregate form in accordance with ethics approval and participant consent, and will be deposited in the Open Science Framework (OSF) under a temporary embargo. raw qualitative data (audio or identifiable transcripts) will remain confidential as required by ethics guidelines. Detailed documentation will accompany all shared materials to ensure transparency and reproducibility.

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the amazon

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Competing Interests

The authors declare that there are no competing interests.

Data Availability Statement

This is a study protocol and does not contain original data. The data management plan and the data collection instrument (questionnaires, interview guides, and focus group guides) are provided as Supporting Information. Upon completion of the study, all anonymized data will be made publicly available in accordance with the plan proposed in the Data Management section.

Abstract

Introduction

Climate change intensifies social, economic, and environmental vulnerabilities in Amazonian rural communities, where dependence on natural resources and limited institutional capacity constrain adaptation. The bioeconomy is often proposed as an

alternative pathway for sustainable development, yet empirical evidence of its role in resilience remains scarce.

Objective

This study protocol presents a mixed methods design to evaluate climate adaptation policies and community resilience strategies in agrarian reform settlements, rural territories created through Brazil's land redistribution program, in Pará, Brazil. Although centered on a regional case, the protocol is structured to generate insights applicable to other socioecologically vulnerable rural contexts in the Amazon and comparable regions worldwide.

Methods

The study will be conducted in 15 municipalities across the three regional superintendencies of the National Institute for Colonization and Agrarian Reform. The design integrates (i) secondary data on socioeconomic, climatic, and policy indicators; (ii) semi-structured interviews with municipal managers (n = 1–3 per municipality); (iii) focus groups with community leaders (n = 8–12 per settlement); and (iv) household surveys (n = 20–50 families per settlement). Primary outcomes include the Income and Livelihood Diversification Index (ILDI), Climate Vulnerability Perception Scale (CVPS), Bioeconomy Engagement Index (BEI), and Reported Adaptive Capacity Index (RACI). Secondary measures include the Adaptation Strategy Diversity Index (ASDI) and a Municipal Climate Engagement Typology (MCET). Quantitative analyses will employ multilevel linear and generalized linear models (LMM/GLMM), while qualitative data will be analyzed through reflexive thematic analysis, with final integration achieved via a triangulation framework.

Expected results

This study will generate a comprehensive framework integrating indicators of vulnerability, adaptive capacity, and bioeconomy engagement across agrarian reform settlements in the Eastern Amazon. By combining household surveys, qualitative fieldwork, and municipal-level policy analysis, the protocol provides a transparent, replicable approach for assessing local climate adaptation processes. The study is designed to inform municipal and state policymakers, support the strengthening of local adaptation initiatives, and contribute evidence relevant to Sustainable Development Goals (SDGs 2, 13, and 15).

Introduction

Climate change represents one of the main threats to sustainable development, especially for vulnerable rural communities in regions of high ecological sensitivity such as the Amazon ¹. These populations face complex challenges in which poverty, limited access to public policy programs, and dependence on fragile production systems interact with environmental impacts, exacerbating their vulnerability ^{2,3}. The intensification of extreme events, including prolonged droughts and atypical floods, has already compromised the livelihoods of communities dependent on family farming and natural resources ⁴.

In this context, the bioeconomy has emerged as a potential pathway to reconcile economic development and environmental conservation, relying on the sustainable use of biodiversity and the valorization of traditional knowledge ⁵. However, its implementation faces critical challenges, including conceptual ambiguity - where differing definitions may or may not generate socio-environmental benefits ⁶ - and the pace of climate change, which

may limit the feasibility of economic model's dependent on ecosystems already under pressure ⁷.

Recent scholarship emphasizes that advancing the Amazonian bioeconomy requires not only technological innovations but also multi-scalar policy changes and institutional support ⁸. From a sustainability transitions perspective, the effective scaling of socio-bioeconomy initiatives depends on improvements in infrastructure, value chains, and social organizations, which remain unevenly developed across region ⁹. Research on non-timber forest products - such as açaí, Brazil nut, and babassu - demonstrates their essential role in linking biodiversity conservation with livelihood generation. Yet, while production, distribution, and economic aspects are relatively well documented, legal, regulatory, and socio-political frameworks remain underexplored ¹⁰. Moreover, the interaction of rural families with forests extends beyond economic considerations, encompassing cultural, subsistence, and spiritual dimensions, which complicates the straightforward promotion of bioeconomy value chains. Together, these insights highlight the need for a comprehensive, context-sensitive approach that integrates economic, ecological, and social perspectives. Parallel to these debates, studies show that Amazonian communities are already devising local adaptation strategies to climate variability and extreme events. Research using a Community - Based Adaptation lens has documented a wide range of coping, adaptive, and transformative responses to extreme floods, despite limited governmental support ¹¹. Other work shows how floodplain residents integrate natural hydrological and ecological processes into their management systems, optimizing ecosystem functioning through traditional knowledge ¹². Similarly, farmers in various regions have adopted new crop varieties, irrigation systems, and strategic land-use practices to enhance resilience. Yet they

continue to face barriers such as water scarcity, high input costs, and insufficient technical assistance, underscoring the importance of institutional support and access to sustainable technologies ¹³.

These findings demonstrate the capacity of rural communities to devise locally grounded adaptation responses, but they also reveal persistent structural challenges, particularly weak integration of local strategies into formal public policies. Importantly, while evidence on adaptation is expanding, a critical gap remains in understanding how these processes unfold in agrarian reform settlements, which represent one of the largest rural populations in the Amazon and sit at the intersection of climate vulnerability, land reform, and bioeconomy development.

Despite growing efforts to assess climate impacts and adaptive responses in Amazonian rural settings, methodological approaches remain fragmented and unevenly applied. This protocol contributes to these ongoing discussions by integrating climate adaptation, resilience, and bioeconomy dimensions into a coherent multilevel framework ¹⁴. This gap constrains the design of effective public policies, which often rely on generic indicators that do not capture the social, ecological, and institutional specificities of vulnerable rural populations. Likewise, although the bioeconomy is frequently promoted as a development pathway, evidence on its effectiveness under rapidly changing climatic conditions remains limited and uneven ¹⁵.

This protocol seeks to address these questions by combining the assessment of climate adaptation and mitigation policies with the analysis of bioeconomy chains in agrarian reform settlements. In Brazil, *agrarian reform settlements* (*assentamentos de reforma agrária*) are rural territories created and regulated by the National Institute for Colonization

and Agrarian Reform (INCRA), where land is redistributed to landless or land-poor families as part of the federal land reform policy. Families receiving plots through this program are commonly referred to as *settled families* or *beneficiary families*.

The results will contribute not only to filling local data gaps¹⁶ but also to advancing global debates on climate adaptation¹⁷, bioeconomy¹⁸, and sustainable development¹⁹, directly engaging with the Sustainable Development Goals (SDG 2 – Zero Hunger, SDG 13 – Climate Action, and SDG 15 – Life on Land).

Materials and Methods

Objectives, hypotheses, and research strategies

This study protocol was developed with the aim of identifying and analyzing public policies and community strategies for climate change mitigation and adaptation in municipalities and rural communities in the Eastern Brazilian Amazon. The specific research objectives are: (i) To map and quantify climate vulnerability through the construction and analysis of secondary indicators of exposure, sensitivity, and adaptive capacity. (ii) To identify, through field research, the perceived effects of climate change, adaptation and mitigation strategies, and the main barriers faced by small-scale family farmers in selected rural settlements. (iii) To map municipal and state-level public policies for addressing climate change, including an analysis of the public budget allocated to these actions. (iv) To analyze the perception, implementation challenges, and articulation of climate policies from the perspective of local public managers.

This is an observational, cross-sectional study. Therefore, the analyses are designed to identify robust statistical associations and to explore potential causal mechanisms, but

they cannot definitively establish causality due to the potential for unmeasured confounding and reverse causality. To enhance transparency and reduce researcher degrees of freedom, we pre-specify a concise set of primary outcomes and hypotheses (detailed in Table 1 and the 'Primary and Secondary Outcomes' section). All other analyses will be clearly labeled as secondary or exploratory

The approach integrates public policy indicators (e.g., existence of mitigation and adaptation plans, climate budget) with community perceptions, allowing for a broader analysis of ongoing impacts and adaptation strategies. The data collected under this protocol aims to test hypotheses based on (i) spatial patterns; (ii) socioeconomic patterns; (iii) demographic patterns; (iv) institutional/political patterns; and (v) bioeconomy/climate patterns.

Table 1 summarizes the main hypotheses guiding this work.

Table 1 – Hypotheses Guiding the Protocol

Analyzed Pattern	Primary Hypothesis (PH)	Theoretical Premise
Spatial and Environmental	PH1: Settlements with greater forest cover and ecological connectivity are expected to be associated with higher adaptive capacity, more diverse of bioeconomy practices, and a lower perception of climate vulnerability.	Intact ecosystems provide more ecosystem services (provisioning, regulating) that function as buffers against climate impacts and as a foundation for resilient economic strategies ^{20, 21, 22}
Socioeconomic and Livelihoods	PH2: Household income diversification (agricultural, extractive, non-agricultural) is positively associated with higher perceived	Diversification reduces dependence on a single activity vulnerable to climate events, enabling the

	climate resilience and greater adoption of adaptation actions.	absorption of shocks and the reallocation of resources ^{23, 24}
Demographic and Social Capital	PH3: Multigenerational demographic composition and participation in community organizations are positively correlated with the adoption of a more diversified portfolio of adaptation strategies (both traditional and innovative).	Intergenerationality facilitates the transmission of traditional knowledge, while social organization facilitates access to information, resources, and external policies ^{25, 26} .
Institutional and Public Policies	PH4: The existence and implementation of participatory public policies on climate and bioeconomy explicitly at the municipal level are associated with higher adaptive capacity reported by communities and greater productive diversification based on socio-biodiversity.	Well-designed and participatory policies provide the resources, incentives, and structures necessary for communities to implement effective adaptation strategies at scale ^{27, 28} .
Bioeconomy and Climate	PH5: Engagement in bioeconomy value chains (particularly processing and certification) is positively associated with higher household income levels and the adoption of climate mitigation and adaptation practices.	Value addition and access to differentiated markets create economic incentives for conservation and for investment in sustainable and resilient practices ^{29, 30} .

182

183 **Primary and Secondary Outcomes**

184 To reduce researcher degrees of freedom and enhance transparency, this protocol pre-
185 specifies four primary outcomes directly linked to the hypotheses presented in Table 1:
186 (i) the Income and Livelihood Diversification Index (ILDI, linked to PH2), (ii) the
187 Climate Vulnerability Perception Scale (CVPS, linked to PH1), (iii) the Bioeconomy

Engagement Index (BEI, linked to PH5), and (iv) the Reported Adaptive Capacity Index (RACI, linked to PH4). All other indicators are classified as secondary and will be used for exploratory analysis. The conceptual definitions are presented below, while the detailed computational procedures and statistical validation strategies are provided in the section *Operationalization of the Indices*.

Based on the hypotheses presented in Table 1, this protocol pre-specifies a set of primary and secondary outcomes to ensure analytical parsimony, clarity, and replicability.

Primary outcomes (core indicators of household resilience and bioeconomy engagement):

1. Income and Livelihood Diversification Index (ILDI): a continuous score (0–1) that synthesizes the distribution of agricultural, extractive, and non-agricultural income. This index is adapted from standard diversification measures widely used in livelihood and rural development research (e.g., Herfindahl–Hirschman approaches³¹). *Expected direction*: higher values will be associated with greater diversification of adaptation actions (PH2).
2. Climate Vulnerability Perception Scale (CVPS): a reflective scale (1–5) constructed from Likert-type items assessing perceived frequency, severity, and impacts of climate-related events. This scale is adapted conceptually from prior work on climate risk perception but operationalized specifically for Amazonian rural contexts. Internal consistency and dimensionality will be assessed through Cronbach’s alpha³² and confirmatory factor analysis. *Expected direction*: lower perceived vulnerability in settlements with greater forest cover and ecological connectivity (PH1).

3. Bioeconomy Engagement Index (BEI): an ordinal variable (level 0 = no engagement; 1 = primary production; 2 = processing; 3 = processing + certification). This is an original indicator developed by the authors, grounded in literature on value-chain upgrading and socio-bioeconomy transitions. *Expected direction*: Higher values reflect greater involvement in bioeconomy value chains. This indicator will be used to examine whether higher engagement is associated with household income and with adoption of adaptation and mitigation practices (PH5).

4. Reported Adaptive Capacity Index (RACI): a reflective score derived from farmers' self-assessment of their ability to respond to climate impacts. The index is constructed as the average of Likert-scale items (range 1–5), where higher values indicate greater perceived adaptive capacity. Items draw on established adaptation assessment frameworks but were adapted to the Amazonian rural context. This indicator is used to examine associations proposed in PH4.

Households with multigenerational demographic composition and participation in community organizations (PH3) will report higher adaptive capacity (as measured by the RACI), and this association will be supported by a more diversified portfolio of adaptation strategies (ASDI). This hypothesis is grounded in demo-livelihoods frameworks³³, which emphasize the role of intergenerational structures and social capital in shaping livelihood decisions and adaptive responses.

Secondary outcomes (exploratory measures to enrich analysis):

a) Adaptation Strategy Diversity Index (ASDI): a count-based index capturing the number and variety of adaptation practices reported by each household, including both

traditional (e.g., collective work, soil management, water storage) and innovative strategies (e.g., improved seeds, irrigation systems, shade management). This is an author-developed indicator, conceptually grounded in livelihood and adaptation portfolio approaches widely used in climate adaptation research. The index reflects the breadth of a household's adaptation repertoire, with higher values indicating a more diversified set of responses to climate impacts. The ASDI supports the examination of PH3 by providing a complementary measure of how demographic composition and social capital relate to adaptation strategy diversity.

b) Municipal Climate Engagement Typology: a contextual variable (high, medium, or low engagement) constructed from a systematic analysis of municipal climate-related policies and budgets. The typology follows consolidated methodologies^{34, 35}, combining two dimensions: (i) *policy engagement*, assessed through the presence, depth, and quality of municipal adaptation and mitigation plans (including objectives, instruments, and implementation mechanisms); and (ii) *budgetary engagement*, assessed through the classification and quantification of climate-relevant expenditures in municipal budgets. Municipalities are then categorized into high, medium, or low engagement based on their combined policy and budget scores (used to test HP4).

All outcomes will be operationalized on continuous, ordinal, or categorical scales, as appropriate, and incorporated into multilevel linear and generalized linear models (LMM/GLMM), with families (level 1) nested within settlements (level 2), which are nested within municipalities (level 3).

A complete description of how each indicator is computed, including variable selection, scaling, aggregation, and validation procedures, is provided in the section Operationalization of the Indices.

Data Collection Methods

The study employs three complementary data collection methods: household surveys, semi-structured interviews, and focus groups. The household survey captures quantitative information on socioeconomic characteristics, productive activities, climate perceptions, and adaptation practices, using a structured questionnaire administered face-to-face to a randomly selected adult decision-maker in each household.

Qualitative data will be collected through two instruments. First, semi-structured interviews will be conducted with municipal public managers (1–3 per municipality), allowing in-depth exploration of policy implementation challenges, resource constraints, and perceptions of local climate risks. Second, focus groups with community leaders (8–12 participants per settlement) will be used to capture collective experiences, shared vulnerabilities, and community-level adaptation strategies. Both methods are designed to generate rich contextual information that complements survey findings.

Detailed guides for each instrument (survey, interview, and focus group) are provided in the Supplementary Material (S1–S3 Files).

Ethical Considerations

This protocol was approved by the Research Ethics Committee of the Federal University of Pará (CAAE: 87492325.6.0000.0018). All participants will provide informed consent

prior to data collection. Identifiable information will be removed during transcription and data processing, and all results will be reported in aggregated form to ensure confidentiality.

Sample, Inclusion and Exclusion Criteria, and Participant Characteristics

The study will be conducted in the state of Pará, in the Eastern Brazilian Amazon. The selection of this state as the study area is strategically grounded in its centrality to debates on agrarian development, environmental conservation, and climate justice in the Brazilian Amazon ³⁶.

Critically, Pará concentrates approximately 25% of all agrarian reform settlements in the country, hosting about 27% of all settled families in Brazil, representing the largest absolute population in agrarian reform projects within a single state. This demographic density of settlements, located in the Amazon biome - which is globally recognized as one of the most climate-sensitive - creates a unique and crucial socioecological context for investigation ³⁷.

Focusing on this state allows the examination, on a microscale, of the challenges and adaptation strategies of a vast and vulnerable population, whose experiences are directly shaped by the intersection of land tenure policies, deforestation pressures, and the intensification of climate extremes. Therefore, the results generated from this sample will not only have significant local relevance but will also provide transferable insights to other regions of the Pan-Amazon facing similar dynamics, directly contributing to filling evidence gaps in climate adaptation and mitigation policies grounded in social equity.

To operationalize research in this unique context, the study will be conducted in ten municipalities of Pará, strategically distributed across the three regional superintendencies of the National Institute for Colonization and Agrarian Reform (INCRA).

The choice of 15 municipalities is not arbitrary but rather designed to capture the diverse socioecological contexts of agrarian reform settlements in Pará. These municipalities encompass distinct territorial and economic features that are central to understanding adaptation dynamics in the Amazon. They include: (i) areas strongly influenced by mining activities, where extractive industries create both opportunities and vulnerabilities; (ii) regions dominated by agribusiness expansion, with intensive land-use change and pressure on family farming; (iii) floodplain and dryland environments, which present contrasting climate risks and livelihood strategies; (iv) municipalities intersected by major highways such as the Transamazon Highway, where infrastructure development shapes access to markets and exposure to environmental impacts; and (v) extractive and conservation-oriented settlements, where socio-biodiversity chains and traditional livelihoods play a central role.

By sampling across this range of contexts, the study captures key socioecological environments characteristic of Amazonian rural settlements. The selection is not numerically balanced across all contexts; rather, municipalities were purposively chosen to ensure representation of major settlement types and territorial dynamics relevant to climate adaptation. This approach enhances analytical robustness and supports the transferability of findings.

In each selected municipality, up to two agrarian reform settlements will be sampled, ensuring representation of the main settlement modalities nationally recognized. This approach will enable a robust comparative analysis of how different institutional designs and production bases mediate community vulnerability and adaptive capacity in the face of climate change.

Selection of settlements

Eligible settlements must: (i) have between 20 and 500 settled families, ensuring comparability and logistical feasibility; (ii) reflect production and socioeconomic conditions representative of the municipality; and (iii) be in a stage of creation, installation, structuring, or consolidation. Settlements with very particular or atypical conditions, such as experimental areas or arrangements that are not replicable, will be excluded.

Participant characteristics

Participants include: (i) local public managers (1–3 per municipality), responsible for municipal policies on agriculture, environment, or rural development; (ii) community leaders (8–12 per settlement), invited to participate in focus groups, representing different social and productive segments; and (iii) settled families (up to 50 per settlement), covering a diversity of socioeconomic, productive, and demographic profiles. The study expects to include family farmers across different age groups, education levels, family compositions, and modes of engagement in local markets, as well as varying degrees of dependence on natural resources and participation in public policies. In each household,

the respondent will be the adult member primarily responsible for agricultural and livelihood decisions (typically the main decision-maker), as this person is expected to have the most comprehensive knowledge of production, income sources, and climate-related impacts.

This sampling strategy seeks to balance territorial representativeness, social diversity, and logistical feasibility, enabling comparisons across municipalities, settlement types, and family profiles.

Selection of families

In each selected settlement, up to 50 families will participate in the study, chosen through simple random sampling from official lists provided by local associations or INCRA. The inclusion criteria for families are: (i) residing in the settlement; and (ii) having an adult household member available to respond to the questionnaire; and (iii) providing voluntary informed consent (ICF). Families will be excluded if: (i) they do not meet the residency requirement; (ii) they refuse or cannot provide consent; or (iii) communication barriers make the interview unfeasible.

Qualitative saturation criterion

Qualitative data collection will follow the principle of theoretical saturation³⁸, assessed iteratively by the analysis team during coding. As a guiding strategy, we will consider saturation reached when two or more consecutive interviews or focus groups no longer generate new relevant codes. However, given the high heterogeneity of Amazonian settlements (e.g., floodplains, mining zones, forest-based communities), this criterion will

be applied flexibly. Data collection may be extended in specific contexts if new themes continue to emerge or if settlement-specific dynamics require further exploration. To ensure robustness, saturation will be evaluated both within settlement types and across municipalities and will be triangulated with survey findings and field observations to confirm completeness.

Sample justification and statistical power

The study was designed to include 15 municipalities, up to 30 settlements, and 600–1,000 families. Power calculations were conducted to determine the minimum sample size needed to achieve at least 80% power at a 5% significance level for detecting modest-to-moderate associations in a multilevel framework. Because key drivers of climate resilience and bioeconomy engagement (e.g., market access, institutional presence, ecological context) operate predominantly at the community level, statistical power is primarily determined by the number of settlements rather than the number of families per settlement.

To reflect this hierarchical structure, simulations incorporated a range of plausible intraclass correlation coefficients (ICCs = 0.05, 0.10, 0.15, 0.30), informed by the literature³⁹ and pilot data. Models were specified to match our primary outcomes: continuous outcomes (e.g., ILDI) were evaluated using linear mixed models (LMM), and ordinal outcomes (e.g., the four-level BEI) using cumulative link mixed models (CLMM). For a sample of 600 families across 30 settlements (average cluster size ≈ 20), $\alpha = 0.05$, and ICC = 0.10, the design provides approximately 80% power to detect a standardized mean difference of 0.35 for continuous outcomes, and to detect an odds

ratio of around 2.0 per one-unit increase in an ordinal predictor for BEI under a proportional-odds structure.

These analyses demonstrate that the planned sample is adequately powered to detect effects of practical relevance for our primary hypotheses. If logistical constraints arise during fieldwork, priority will be given to maximizing the number of settlements rather than increasing the number of families per settlement, as this yields greater power for between-cluster comparisons. Full simulation scripts will be made available in the project's OSF repository to ensure computational transparency and reproducibility³⁹.

Description of All Processes, Interventions, and Comparisons

Data collection will be carried out sequentially in two integrated phases, as detailed below. The methodological design was refined following a pilot study conducted in the municipality of Marabá (PA), with the resulting adjustments described to ensure process transparency.

Phase 1: Secondary Data and Political-Institutional Context Analysis

This phase will focus on building a comprehensive diagnosis of the municipal context. It will involve, first, Climate Risk Assessment, through the compilation and analysis of secondary climate indicators widely recognized in the literature (e.g., temperature, precipitation, extreme events) from sources such as the National Institute of Meteorology (INMET) and the National Institute for Space Research (INPE), to characterize the exposure and sensitivity of the sampled municipalities. Second, Public Policy Mapping, through the identification and documentary analysis of municipal and state-level public

policies related to climate change, agriculture, and rural development. This will include the analysis of master plans, laws, and decrees, with attention to both actions explicitly labeled as “climate-related” and sectoral actions with indirect impacts on adaptation and mitigation (e.g., sanitation, civil defense, family farming) ²⁷. Third, Public Budget Audit, applying a data mining methodology to municipal transparency portals ^{34, 35}, which allows categorization and quantification of public expenditures directly or indirectly related to the climate agenda, thus overcoming the limitation of expenditures formally tagged as “climatic” ⁴⁰.

Phase 2: Primary Data Collection with Key Stakeholders

This phase will capture the perceptions, experiences, and strategies of local actors through three instruments: (i) Household Surveys: administration of a structured questionnaire to a sample of families in each settlement, collecting socioeconomic, productive, climate perception, and adaptation strategy data (S1 File – Household Survey); (ii) Focus Groups with Community Leaders in each settlement type: group discussions to obtain a consolidated view of vulnerabilities, conflicts, and collective adaptation strategies in the settlements, as well as to validate contextual information (S3 File – Focus Group Guide); (iii) Semi-Structured Interviews with Public Managers: interviews with municipal technicians and secretaries to understand perceptions, implementation challenges, and the articulation of climate policies from the perspective of local government (S3 File – Interview Guide).

Post-Pilot Methodological Adjustments.

Testing the instruments in Marabá (PA) resulted in crucial refinements. For Phase 1, the policy and budget analysis methodology was expanded to systematically capture implicit climate actions, as the pilot revealed that most municipal climate responses are driven by sectoral policies (e.g., agriculture, infrastructure) not explicitly labeled as climate related. For Phase 2, the household survey was adapted to include specific modules for riverine and floodplain communities, capturing climate impacts (e.g., changes in flood cycles, river-based access). The interview guide with managers was revised to replace technical jargon with more accessible terms, reflecting the fact that many municipalities lack a consolidated formal framework for climate policy.

Comparative Analysis Strategy

The analysis will be structured across four comparative levels, allowing the isolation of factors associated with resilience at different scales: (i) Across municipalities, contrasting indicators of vulnerability, climate budget, and policy robustness; (ii) Across settlement types, examining how different land tenure and institutional arrangements mediate convergences of families within settlements, correlating socioeconomic profiles (education, income diversification) with adoption of adaptation practices; (iv) Across social actors, triangulating the perspectives of public managers and communities to identify convergences and divergences in risk perceptions and policy effectiveness.

This multilevel approach is essential for identifying robust spatial, socioeconomic, and institutional patterns associated with climate resilience.

Data Analysis Plan

Data analysis will follow a sequential explanatory mixed-methods strategy (QUAN → QUAL)⁴¹, in which the initial quantitative analysis will provide an overall picture to be subsequently deepened and contextualized by qualitative investigation. Triangulation of data from different sources (quantitative, qualitative, and secondary) will be employed to validate and enrich the findings, ensuring a robust understanding of the factors that influence climate resilience⁴². All analyses will be performed using R software (version 4.3.0 or higher) for the quantitative component and NVivo software (version 14 or higher) to support the organization and coding of qualitative data.

General Strategy and Integration

The analysis will occur in three main stages, corresponding to the data sets, culminating in an integration phase: (i) Secondary Data and Municipal Context Analysis, to characterize the macro context. (ii) Quantitative Primary Data Analysis (Survey), aimed at identifying patterns and testing hypotheses at household and community levels. (iii) Qualitative Primary Data Analysis (Interviews and Focus Groups), to explore causal mechanisms, perceptions, and nuances. (iv) Finally, integration and triangulation, with a synthesis of the findings to answer the research objectives.

Secondary Data and Municipal Context Analysis

The analysis of secondary data will proceed in three integrated steps to construct key municipal-level variables. First, a Municipal Climate Vulnerability Index (MCVI) will be created by compiling secondary indicators of exposure, sensitivity, and adaptive capacity from sources including AdaptaBrasil, MCTI, IBGE, and INPE. These indicators will be

483 harmonized by converting to per capita measures where appropriate, standardized (z-
484 scores), and aggregated through Principal Component Analysis (PCA) ⁴³. Second,
485 municipal climate change plans will be assessed using categorical content analysis to
486 generate a plan quality score. Third, a public expenditure audit, employing text mining of
487 municipal transparency portals, will be conducted to identify and quantify climate-related
488 budgets.

489 To ensure robustness, a multi-step strategy will address data quality challenges. For the
490 MCVI, discrepancies or missing values will be handled using imputation techniques, with
491 sensitivity analyses conducted to assess the impact on the index. Furthermore, all
492 secondary data - including the budget analysis and plan documentation - will be cross-
493 checked against municipal transparency portals and triangulated with qualitative data
494 from interviews and focus groups to ensure contextual accuracy.

495 Finally, the plan quality scores, and budget allocations will be combined to create a
496 Municipal Climate Engagement Typology (e.g., High, Medium, Low). Both the MCVI
497 and this engagement typology will be incorporated as contextual variables in the
498 multilevel models.

500 **Quantitative Primary Data Analysis (Survey)**

501 Initially, descriptive statistics (frequencies, means, medians, and standard deviations) will
502 be used to characterize the sample profile. Subsequently, to operate key concepts,
503 composite indices will be constructed, and their internal consistency will be assessed
504 using Cronbach's alpha ($\alpha > 0.7$) and/or Confirmatory Factor Analysis (CFA) ⁴⁴. The
505 planned indices include Income and Livelihood Diversification Index (ILDI); Climate

Adaptation Practices Index (CAPI); Bioeconomy Engagement Index (BEI); and Climate Vulnerability Perception Scale (CVPS).

To test the hypotheses in Table 1, linear mixed-effects models (LMM) and generalized linear mixed-effects models (GLMM)⁴⁵ will be applied, which are appropriate for hierarchical data (families nested within settlements and municipalities). The MCVI and the municipal engagement typology will be included as higher-level variables (level 2), while family characteristics (level 1) will be used as predictors and control variables.

Addressing Confounding and Sensitivity Analysis

To mitigate potential confounding, the multilevel models will include a rich set of pre-specified covariates at different levels: (1) Household Level (Level 1): Age and education of the household head, household size, gender of the household head, asset index, length of residence in the settlement; (2) Settlement Level (Level 2): Settlement type, distance to the nearest paved road, presence of a producers' cooperative, years since settlement establishment; (3) Municipality Level (Level 3): Municipal Climate Vulnerability Index (MCVI), Municipal Climate Engagement Typology, and geographic region (e.g., floodplain vs. upland)

Acknowledging the impossibility of measuring all potential confounders, we will conduct a formal sensitivity analysis for the key associations related to our primary hypotheses (e.g., the association between BEI and household income). We will use the `sensemakr` package in R, following the framework developed by Cinelli and Hazlett (2025)⁴⁶. For each key finding, we will report the Robustness Value (RV), the minimum strength of association (partial R^2) that an unmeasured confounder would need to have with both the

exposure and the outcome to explain away the observed effect. We will also present contour plots to visualize how the effect estimate and its confidence interval would change under different scenarios of confounding. This will provide readers with a transparent assessment of the robustness of our inferences to unobserved confounding.

Operationalization of the Indices

To ensure transparency and replicability, each index will be constructed according to clear theoretical and statistical criteria, as follows:

- Income and Livelihood Diversification Index (ILDI): formative index calculated using the Herfindahl-Hirschman formula $(1 - \sum p_i^2)^{31}$ based on the relative shares of agricultural, extractive, and non-agricultural income. It will not be evaluated by Cronbach's alpha but described statistically.
- Climate Adaptation Practices Index (CAPI): reflective index, constructed from the standardized sum of reported adaptation practices (e.g., irrigation, resistant varieties, collective work). Internal consistency will be assessed by Cronbach's alpha³² ($\alpha \geq 0.7$) and, when appropriate, by confirmatory factor analysis.
- Bioeconomy Engagement Index (BEI): formative ordinal variable, with increasing levels of engagement in bioeconomy value chains (0: no participation; 1: primary production; 2: processing; 3: processing + certification/market differentiation). Higher values indicate greater household-level involvement. For multilevel analyses, the BEI will be used primarily as a level-1 predictor (household-level) to examine associations with income, adaptive capacity, and adoption of mitigation/adaptation practices. To explore community-level patterns,

the BEI will also be summarized at the settlement level using the mean or distribution of household scores (e.g., proportion of households at each engagement level). These aggregated measures will be used as level-2 contextual variables when relevant for hypotheses concerning settlement-level dynamics (e.g., HP5).

- Climate Vulnerability Perception Scale (CVPS): reflective scale, constructed as the average of Likert-type items (1–5) capturing perceptions of climate risk (drought, flood, crop loss, food insecurity). Internal consistency will be assessed by Cronbach’s alpha ($\alpha \geq 0.7$) and unidimensionality by CFA.
- Municipal Climate Vulnerability Index (MCVI): formative index based on indicators of exposure (climate), sensitivity (e.g., agricultural dependence, poverty), and adaptive capacity (e.g., education, policy access). It will be constructed from standardized indicators (z-scores) and aggregated through PCA. PCA adequacy indicators ($KMO > 0.6$) and variance explained by the first component will be reported.

Each index will be statistically validated, described in terms of distribution, central tendency, and dispersion, and interpreted according to its reflective or formative nature.

Qualitative Primary Data Analysis

Transcriptions of interviews and focus groups will be analyzed using Braun and Clarke’s Reflexive Thematic Analysis⁴⁷ in six stages: (1) familiarization; (2) generation of initial codes; (3) search for themes; (4) review of themes; (5) definition and naming of themes; and (6) report production. This analysis will identify central themes related to local

perceptions, barriers, facilitators, and understandings of climate change, adaptation, and bioeconomy.

All qualitative findings will be reported according to the COREQ guideline (or SRQR, where applicable), ensuring comprehensive documentation of data collection and analytic procedures.

Integration and Triangulation

In the final phase, the findings will be integrated following a sequential explanatory mixed-methods design (QUAN → QUAL) and a joint display triangulation framework ⁴².

The integration will be reported in accordance with the Good Reporting of a Mixed Methods Study (GRAMMS) framework and the Standards for Reporting Qualitative Research (SRQR) guidelines for the qualitative component.

A triangulation matrix will be constructed to systematically map convergences, complementarities, and divergences between quantitative, qualitative, and secondary data findings for each research objective and hypothesis. The matrix will define decision rules for interpreting relationships: (I) Convergence: Findings from different methods provide corroborating evidence for the same conclusion; (II) Complementarity: Findings from different methods elucidate distinct, non-overlapping aspects of a phenomenon, together providing a richer, more complete picture; (III) Discordance: Findings from different methods are contradictory or inconsistent. In such cases, we will re-examine the data and the assumptions of each method to propose potential explanations for the discrepancy (e.g., differences in perspective between managers and farmers, social desirability bias in

surveys). (IV) Silence: An expected finding from one method is absent in the data from another method

For example, a quantitative pattern of low adoption of a bioeconomy practice (QUAN) will be examined considering reported barriers in interviews (QUAL) and the (non)existence of supportive policies identified in secondary data. This synthesis will enable not only the identification of *what* occurs but also *why* and *how* adaptation and resilience processes unfold.

Handling of Missing Data

The strategy for handling missing data will be determined by its mechanism and extent. During data collection, digital forms with mandatory field checks will minimize inadvertent missingness. Interviewers will be trained to probe for reluctant or "don't know" responses. Prior to analysis, we will generate a missingness pattern report. Little's Missing Completely at Random (MCAR)⁴⁸ test will be used to inform the choice of handling method, though we will primarily assume data is Missing at Random (MAR). Our primary analysis will use a Full Information Maximum Likelihood (FIML) estimation within the mixed-effects models (lme4 package in R). FIML uses all available data points without discarding partial cases and provides unbiased estimates under the Missing at Random (MAR) assumption.

To assess the robustness, we will conduct sensitivity analyses using Multiple Imputation (mice package in R). We will create 20 imputed datasets, including all model variables and auxiliary variables correlated with missingness. The multilevel structure will be respected during imputation using the 2lonly method, incorporating cluster means of

level 1 variables as predictors. All model variables, plus auxiliary variables correlated with missingness (e.g., interview duration, interviewer ID), will be included in the imputation model. Results from the pooled imputed models will be compared with the primary FIML analysis.

In addition to primary data, we will assess and report missingness in all secondary datasets, including climate indicators, socioeconomic data, municipal plans, and budget information. For missing values in external data sources, we will apply appropriate strategies depending on the mechanism: (i) triangulation with alternative official databases when equivalent indicators exist, (ii) imputation of continuous variables using multilevel or time-series-based methods when justified, and (iii) explicit coding of structural missingness (e.g., municipalities with no climate plans or no identifiable climate-related expenditures). The extent and implications of missing secondary data will be documented in the Municipal Climate Vulnerability Index (MCVI) and in the construction of the Municipal Climate Engagement Typology (MCET), and sensitivity checks will be conducted to assess the robustness of results to these gaps

Data Management Plan

All secondary data comes from public sources and will be fully documented. Primary surveys and qualitative data will undergo rigorous anonymization to protect participant confidentiality. Anonymized datasets, metadata, and analysis scripts will be deposited in the Open Science Framework (OSF) under a temporary embargo; raw qualitative data (audio or identifiable transcripts) will remain confidential as required by ethics

guidelines. Detailed documentation will accompany all shared materials to ensure transparency and reproducibility.

Study Status and Timeline

A pilot study was conducted in Marabá (PA) to refine instruments and procedures. Large-scale data collection is scheduled for December 2025 to December 2026, followed by data analysis and dissemination activities in 2027. These timelines reflect logistical constraints of fieldwork in remote Amazonian settlements.

Discussion

Study Limitations

The main methodological advancement of this protocol lies in the integration of climate resilience indicators with bioeconomy dynamics in agrarian reform settlements. To our knowledge, this approach has not been systematically applied in international studies, reflecting both the novelty and global relevance of this work. Nevertheless, several limitations should be acknowledged, together with justifications for the chosen design. First, regarding study design, the mixed-methods and multilevel analytical approach was selected because climate adaptation and bioeconomy practices are shaped simultaneously by household characteristics, community organization, and municipal institutional capacity. A purely quantitative design would not capture the mechanisms, narratives, and context-specific meanings that influence adaptive behavior, whereas a purely qualitative design would not allow systematic comparison across municipalities and settlements. The

664 sequential explanatory design (QUAN → QUAL), supported by triangulation, therefore
665 provides an appropriate structure for addressing the research questions. Alternative
666 designs were considered, particularly a longitudinal panel design and an experimental or
667 quasi-experimental evaluation of policy interventions but were not feasible due to the
668 absence of comparable baseline data, the difficulty of randomization in this context, and
669 the logistical challenges of multi-year fieldwork in remote Amazonian territories.

670 Second, the reliance on secondary data introduces constraints related to data availability,
671 variable quality, and inconsistent reporting across municipalities. Climate indicators,
672 budget classifications, and policy documents vary in completeness and temporal
673 coverage. To mitigate these limitations, the study incorporates (i) triangulation across
674 multiple official databases, (ii) imputation and sensitivity analyses for incomplete
675 indicators, and (iii) explicit treatment of structural missingness (e.g., municipalities with
676 no climate plans or no identifiable climate-related expenditures).

677 Third, the qualitative component may be vulnerable to social desirability and
678 positionality biases. Interviewers may provide responses they perceive as desirable or
679 aligned with institutional expectations. This risk will be minimized through careful
680 interviewer training, rapport-building, and emphasis on confidentiality and the absence of
681 “right” or “wrong” answers. Additionally, by drawing on community-based researchers
682 familiar with the local context, the study enhances trust and reduces outsider bias.

683 Fourth, the intentional selection of municipalities and settlements limits the
684 generalizability of findings for the entire state of Pará. The study does not aim to produce
685 a statistically representative estimate of all settlements; instead, it provides an in-depth
686 analysis of key socioecological contexts. However, the detailed methodological

687 description and the diversity of selected municipalities allow assessment of the
688 transferability of insights to similar contexts across the Eastern Amazon.

689 Fifth, although the protocol draws on international literature on climate adaptation,
690 resilience, and rural livelihoods, it extends this body of work in important ways. Previous
691 studies often examine adaptation or bioeconomy pathways separately; few integrate both
692 dimensions with a multilevel analytical structure and grounded qualitative evidence. The
693 protocol also adapts global frameworks (e.g., livelihood diversification, adaptation
694 portfolios, municipal climate governance) to the institutional realities of agrarian reform
695 settlements, an understudied but globally relevant category of rural territories.

696 Finally, the potential applications of the findings justify the chosen approach. Once
697 completed, the study will produce evidence directly applicable to municipal and state
698 policymakers in Brazil, supporting the refinement of climate adaptation and bioeconomy
699 strategies, the allocation of public resources, and the design of territorial programs.

700 Internationally, the protocol can inform studies in other regions with decentralized
701 climate governance, rural poverty, and socio-biodiversity economies, such as Southeast
702 Asia, Central America, and parts of Sub-Saharan Africa. With minor contextual
703 adaptations (e.g., indicator selection, policy frameworks, and institutional architecture),
704 the methodological approach can be generalized to rural areas undergoing similar
705 socioecological transitions.

706 Overall, while the study faces limitations inherent to its observational and context-
707 specific design, the mixed-methods, multilevel structure, combined with transparency,
708 triangulation, and sensitivity analyses, provides a robust framework for advancing
709 empirical knowledge on climate resilience and bioeconomy pathways in vulnerable rural

regions. In the final analysis, the implications of each limitation will be explicitly assessed through sensitivity analyses, triangulation, and transparent reporting of uncertainty, and these will be clearly communicated in the presentation of results to ensure that conclusions accurately reflect the strength and boundaries of the evidence.

Author's Contributions

Conceptualization: Daniel Silva.

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Funding acquisition: Daniel Silva.

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Supporting Information

S1 File. Focal Groups Guide

S2 File. Interview Guide for Public Managers

S3 File. Questionnaire Guide for Family Farmers

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Supporting Information - Compressed/ZIP File Archive
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Climate resilience through bioeconomy: A mixed-methods
protocol for assessing adaptation policies in rural settlements on
the amazon

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39
40 **Competing Interests**

41 The authors declare that there are no competing interests.
42

43 **Data Availability Statement**

44 This is a study protocol and does not contain original data. The data management plan
45 and the data collection instrument (questionnaires, interview guides, and focus group
46 guides) are provided as Supporting Information. Upon completion of the study, all
47 anonymized data will be made publicly available in accordance with the plan proposed in
48 the Data Management section.
49

50 **Abstract**

51 **Introduction**

52 Climate change intensifies social, economic, and environmental vulnerabilities in
53 Amazonian rural communities, where dependence on natural resources and limited

institutional capacity constrain adaptation. The bioeconomy is often proposed as an alternative pathway for sustainable development, yet empirical evidence of its role in resilience remains scarce.

Objective

~~This protocol outlines a mixed methods study designed to evaluate public policies for climate adaptation and to analyze community resilience strategies in agrarian reform settlements in Pará, Brazil.~~

This study protocol presents a mixed methods design to evaluate climate adaptation policies and community resilience strategies in agrarian reform settlements, rural territories created through Brazil's land redistribution program, in Pará, Brazil. Although centered on a regional case, the protocol is structured to generate insights applicable to other socioecologically vulnerable rural contexts in the Amazon and comparable regions worldwide.

Methods

The study will be conducted in ~~40–15~~ municipalities across the three regional superintendencies of the National Institute for Colonization and Agrarian Reform. The design integrates (i) secondary data on socioeconomic, climatic, and policy indicators; (ii) semi-structured interviews with municipal managers (n = 1–3 per municipality); (iii) focus groups with community leaders (n = 8–12 per settlement); and (iv) household surveys (n = 20–50 families per settlement). Primary outcomes include the Income and Livelihood Diversification Index (ILDI), Climate Vulnerability Perception Scale (CVPS), Bioeconomy Engagement Index (BEI), and Reported Adaptive Capacity Index (RACI).

Secondary measures include the Adaptation Strategy Diversity Index (ASDI) and a Municipal Climate Engagement Typology (MCET). Quantitative analyses will employ multilevel linear and generalized linear models (LMM/GLMM), while qualitative data will be analyzed through reflexive thematic analysis, with final integration achieved via a triangulation framework.

Expected results

~~The study will generate integrated indicators of vulnerability, adaptive capacity, and bioeconomy engagement, enabling the identification of spatial, socioeconomic, and institutional patterns associated with resilience. Findings are expected to inform municipal and state policymakers, strengthen local adaptation initiatives, and contribute to the achievement of Sustainable Development Goals (SDGs 2, 13, and 15).~~

This study will generate a comprehensive framework integrating indicators of vulnerability, adaptive capacity, and bioeconomy engagement across agrarian reform settlements in the Eastern Amazon. By combining household surveys, qualitative fieldwork, and municipal-level policy analysis, the protocol provides a transparent, replicable approach for assessing local climate adaptation processes. The study is designed to inform municipal and state policymakers, support the strengthening of local adaptation initiatives, and contribute evidence relevant to Sustainable Development Goals (SDGs 2, 13, and 15).

Introduction

Climate change represents one of the main threats to sustainable development, especially for vulnerable rural communities in regions of high ecological sensitivity such as the

Amazon ¹. These populations face complex challenges in which poverty, limited access to public ~~policy programs~~^{policies}, and dependence on fragile production systems interact with environmental impacts, exacerbating their vulnerability ^{2, 3}. The intensification of extreme events, including prolonged droughts and atypical floods, has already compromised the livelihoods of communities dependent on family farming and natural resources ⁴.

In this context, the bioeconomy has emerged as a potential pathway to reconcile economic development and environmental conservation, relying on the sustainable use of biodiversity and the valorization of traditional knowledge ⁵. However, its implementation faces critical challenges, including conceptual ambiguity - where differing definitions may or may not generate socio-environmental benefits ⁶ - and the pace of climate change, which may limit the feasibility of economic models dependent on ecosystems already under pressure ⁷.

Recent scholarship emphasizes that advancing the Amazonian bioeconomy requires not only technological innovations but also multi-scalar policy changes and institutional support ⁸. From a sustainability transitions perspective, the effective scaling of socio-bioeconomy initiatives depends on improvements in infrastructure, value chains, and social organizations, which remain unevenly developed across region ⁹. Research on non-timber forest products (~~NTFPs~~)— such as açai, Brazil nut, and babassu - demonstrates their essential role in linking biodiversity conservation with livelihood generation. Yet, while production, distribution, and economic aspects are relatively well documented, legal, regulatory, and socio-political frameworks remain underexplored ¹⁰. Moreover, ~~the interaction of~~ rural families' ~~interactions~~ with forests extends beyond economic

123 considerations, encompassing cultural, subsistence, and spiritual dimensions, which
124 complicates the straightforward promotion of bioeconomy value chains. Together, these
125 insights highlight the need for a comprehensive, context-sensitive approach that integrates
126 economic, ecological, and social perspectives.

127 Parallel to these debates, studies show that Amazonian communities are already devising
128 local adaptation strategies to climate variability and extreme events. Research using a
129 Community-Based Adaptation (CBA) lens has documented a wide range of coping,
130 adaptive, and transformative responses to extreme floods, despite limited governmental
131 support ¹¹. Other work shows how floodplain residents integrate natural hydrological and
132 ecological processes into their management systems, optimizing ecosystem functioning
133 through traditional knowledge ¹². Similarly, farmers in various regions have adopted new
134 crop varieties, irrigation systems, and strategic land-use practices to enhance resilience.
135 Yet they continue to face barriers such as water scarcity, high input costs, and insufficient
136 technical assistance, underscoring the importance of institutional support and access to
137 sustainable technologies ¹³.

138 These findings demonstrate the capacity of rural communities to devise locally grounded
139 adaptation responses, but they also reveal persistent structural challenges, particularly
140 weak integration of local strategies into formal public policies. Importantly, while evidence
141 on adaptation is expanding, a critical gap remains in understanding how these processes
142 unfold in agrarian reform settlements, which represent one of the largest rural populations
143 in the Amazon and sit at the intersection of climate vulnerability, land reform, and
144 bioeconomy development.

145 Despite growing efforts to assess climate impacts and adaptive responses in Amazonian
146 rural settings, methodological approaches remain fragmented and unevenly applied. This
147 protocol contributes to these ongoing discussions by integrating climate adaptation,
148 resilience, and bioeconomy dimensions into a coherent multilevel framework~~Despite~~
149 ~~recognition of the importance of climate adaptation, there is still no standardized~~
150 ~~methodology to assess local impacts and adaptive responses in Amazonian rural contexts~~
151 ^{14.} This gap constrains the design of effective public policies, which often rely on generic
152 indicators that do not capture the social, ecological, and institutional specificities of
153 vulnerable rural populations. Likewise, although the bioeconomy is frequently promoted
154 as a development pathway, evidence on its effectiveness under rapidly changing climatic
155 conditions remains limited and uneven ~~This limits the formulation of effective public~~
156 ~~policies, which are often based on generic indicators that fail to reflect the specific realities~~
157 ~~of vulnerable populations. Similarly, bioeconomy initiatives, although widely promoted,~~
158 ~~still lack robust evidence of their effectiveness in scenarios of accelerated climate change~~
159 ^{15.}

160 This protocol seeks to address these ~~gaps-questions~~ by combining the assessment of climate
161 adaptation and mitigation policies with the analysis of bioeconomy chains in agrarian
162 reform settlements. In Brazil, agrarian reform settlements (*assentamentos de reforma*
163 *agrária*) are rural territories created and regulated by the National Institute for Colonization
164 and Agrarian Reform (INCRA), where land is redistributed to landless or land-poor
165 families as part of the federal land reform policy. Families receiving plots through this
166 program are commonly referred to as *settled families* or *beneficiary families*.

~~To our knowledge, this represents an unprecedented perspective in the international literature on adaptation in tropical regions.~~ The results will contribute not only to filling local data gaps¹⁶ but also to advancing global debates on climate adaptation¹⁷, bioeconomy¹⁸, and sustainable development¹⁹, directly engaging with the Sustainable Development Goals (SDG 2 – Zero Hunger, SDG 13 – Climate Action, and SDG 15 – Life on Land).

Materials and Methods

Objectives, hypotheses, and research strategies

This study protocol was developed with the aim of identifying and analyzing public policies and community strategies for climate change mitigation and adaptation in municipalities and rural communities in the Eastern Brazilian Amazon. The specific research objectives are: (i) To map and quantify climate vulnerability through the construction and analysis of secondary indicators of exposure, sensitivity, and adaptive capacity. (ii) To identify, through field research, the perceived effects of climate change, adaptation and mitigation strategies, and the main barriers faced by small-scale family farmers in selected rural settlements. (iii) To map municipal and state-level public policies for addressing climate change, including an analysis of the public budget allocated to these actions. (iv) To analyze the perception, implementation challenges, and articulation of climate policies from the perspective of local public managers.

This is an observational, cross-sectional study. Therefore, the analyses are designed to identify robust statistical associations and to explore potential causal mechanisms, but they cannot definitively establish causality due to the potential for unmeasured

confounding and reverse causality. To enhance transparency and reduce researcher degrees of freedom, we pre-specify a concise set of primary outcomes and hypotheses (detailed in Table 1 and the 'Primary and Secondary Outcomes' section). All other analyses will be clearly labeled as secondary or exploratory

The approach integrates public policy indicators (e.g., existence of mitigation and adaptation plans, climate budget) with community perceptions, allowing for a broader analysis of ongoing impacts and adaptation strategies. The data collected under this protocol aims to test hypotheses based on (i) spatial patterns; (ii) socioeconomic patterns; (iii) demographic patterns; (iv) institutional/political patterns; and (v) bioeconomy/climate patterns.

Table 1 summarizes the main hypotheses guiding this work.

Table 1 – Hypotheses Guiding the Protocol

Analyzed Pattern	Primary Hypothesis (PH)	Theoretical Premise
Spatial and Environmental	PH1: Settlements with greater forest cover and ecological connectivity <u>are expected to be associated with</u> higher adaptive capacity, <u>more diverse reflected in a greater diversity</u> of bioeconomy practices, and a lower perception of climate vulnerability.	Intact ecosystems provide more ecosystem services (provisioning, regulating) that function as buffers against climate impacts and as a foundation for resilient economic strategies ^{20, 21, 22}
Socioeconomic and Livelihoods	PH2: Household income diversification (agricultural, extractive, non-agricultural) is positively associated with higher perceived climate resilience and greater adoption of adaptation actions.	Diversification reduces dependence on a single activity vulnerable to climate events, enabling the absorption of shocks and the reallocation of resources ^{23, 24}

Demographic and Social Capital	PH3: Multigenerational demographic composition and participation in community organizations are positively correlated with the adoption of a more diversified portfolio of adaptation strategies (both traditional and innovative).	Intergenerationality facilitates the transmission of traditional knowledge, while social organization facilitates access to information, resources, and external policies ^{25, 26} .
Institutional and Public Policies	PH4: The existence and implementation of participatory public policies on climate and bioeconomy explicitly at the municipal level are associated with higher adaptive capacity reported by communities and greater productive diversification based on socio-biodiversity.	Well-designed and participatory policies provide the resources, incentives, and structures necessary for communities to implement effective adaptation strategies at scale ^{27, 28} .
Bioeconomy and Climate	PH5: Engagement in bioeconomy value chains (particularly processing and certification) is positively associated with higher household income levels and the adoption of climate mitigation and adaptation practices.	Value addition and access to differentiated markets create economic incentives for conservation and for investment in sustainable and resilient practices ^{29, 30} .

Primary and Secondary Outcomes

To reduce researcher degrees of freedom and enhance transparency, this protocol pre-specifies four primary outcomes directly linked to the hypotheses presented in Table 1: (i) the Income and Livelihood Diversification Index (ILDI, linked to PH2), (ii) the Climate Vulnerability Perception Scale (CVPS, linked to PH1), (iii) the Bioeconomy Engagement Index (BEI, linked to PH5), and (iv) the Reported Adaptive Capacity Index

(RACI, linked to PH4). All other indicators are classified as secondary and will be used for exploratory analysis. The conceptual definitions are presented below, while the detailed computational procedures and statistical validation strategies are provided in the section *Operationalization of the Indices*.

Based on the hypotheses presented in Table 1, this protocol pre-specifies a set of primary and secondary outcomes to ensure analytical parsimony, clarity, and replicability.

Primary outcomes (core indicators of household resilience and bioeconomy engagement):

1. Income and Livelihood Diversification Index (ILDI): a continuous score (0–1) that synthesizes the distribution of agricultural, extractive, and non-agricultural income. This index is adapted from standard diversification measures widely used in livelihood and rural development research (e.g., Herfindahl–Hirschman approaches³¹). *Expected direction:* higher values will be associated with greater ~~adoption~~ diversification of adaptation actions (PH2).
2. Climate Vulnerability Perception Scale (CVPS): ~~a reflective scale~~ an average score (1–5) constructed from Likert-type items assessing perceived frequency, severity, and impacts of climate-related events ~~measuring climate risk perception~~. This scale is adapted conceptually from prior work on climate risk perception but operationalized specifically for Amazonian rural contexts. Internal consistency and dimensionality will be assessed through Cronbach’s alpha³² and confirmatory factor analysis. *Expected direction:* lower perceived vulnerability in settlements with greater forest cover and ecological connectivity (PH1).
3. Bioeconomy Engagement Index (BEI): an ordinal variable (level 0 ~~=~~ = no engagement; 1 ~~=~~ = primary production; 2 ~~=~~ = processing; 3 ~~=~~ = processing +

certification). This is an original indicator developed by the authors, grounded in literature on value-chain upgrading and socio-bioeconomy transitions. Expected direction: Higher values reflect greater involvement in bioeconomy value chains. This indicator will be used to examine whether higher engagement is associated ~~with greater engagement will be associated with higher average~~ household income and with adoption of adaptation and mitigation practices (PH5).

4. Reported Adaptive Capacity Index (RACI): a reflective score derived from farmers' self-assessment of their ability to respond to climate impacts. The index is constructed as the average of Likert-scale items (range 1–5), where higher values indicate greater perceived adaptive capacity. Items draw on established adaptation assessment frameworks but were adapted to the Amazonian rural context. This indicator is used to examine associations proposed in events, used to test PH4.

Households with multigenerational demographic composition and participation in community organizations (PH3) will report higher adaptive capacity (as measured by the RACI), and this association will be supported by a more diversified portfolio of adaptation strategies (ASDI). This hypothesis is grounded in demo-livelihoods frameworks³³, which emphasize the role of intergenerational structures and social capital in shaping livelihood decisions and adaptive responses.

Secondary outcomes (exploratory measures to enrich analysis):

- a) Adaptation Strategy Diversity Index (ASDI): a count-based index capturing the number and variety of adaptation practices reported by each household, including both traditional (e.g., collective work, soil management, water storage) and innovative

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strategies (e.g., improved seeds, irrigation systems, shade management). This is an
 author-developed indicator, conceptually grounded in livelihood and adaptation portfolio
 approaches widely used in climate adaptation research. The index reflects the breadth of a
 household's adaptation repertoire, with higher values indicating a more diversified set of
 responses to climate impacts. The ASDI supports the examination of PH3 by providing a
 complementary measure of how demographic composition and social capital relate to
 adaptation strategy diversity. ~~a count of reported practices, used to test HP3.~~
 b) Municipal Climate Engagement Typology: a contextual variable (high, medium, or
low engagement) constructed from a systematic analysis of municipal climate-related
policies and budgets. The typology follows consolidated methodologies^{34, 35}, combining
two dimensions: (i) *policy engagement*, assessed through the presence, depth, and quality
of municipal adaptation and mitigation plans (including objectives, instruments, and
implementation mechanisms); and (ii) *budgetary engagement*, assessed through the
classification and quantification of climate-relevant expenditures in municipal budgets.
Municipalities are then categorized into high, medium, or low engagement based on their
combined policy and budget scores (used to test HP4). ~~a contextual variable (high,~~
~~medium, or low engagement) constructed from the analysis of municipal plans and public~~
~~budgets (HP4).~~
 All outcomes will be operationalized on continuous, ordinal, or categorical scales, as
 appropriate, and incorporated into multilevel linear and generalized linear models
~~multilevel models~~ (LMM/GLMM), with families (level 1) nested within settlements
(level 2), which are nested within municipalities (level 3) ~~with families (level 1) nested~~
~~within settlements and municipalities (level 2).~~

A complete description of how each indicator is computed, including variable selection, scaling, aggregation, and validation procedures, is provided in the section Operationalization of the Indices.

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Data Collection Methods

The study employs three complementary data collection methods: household surveys, semi-structured interviews, and focus groups. The household survey captures quantitative information on socioeconomic characteristics, productive activities, climate perceptions, and adaptation practices, using a structured questionnaire administered face-to-face to a randomly selected adult decision-maker in each household. Qualitative data will be collected through two instruments. First, semi-structured interviews will be conducted with municipal public managers (1–3 per municipality), allowing in-depth exploration of policy implementation challenges, resource constraints, and perceptions of local climate risks. Second, focus groups with community leaders (8–12 participants per settlement) will be used to capture collective experiences, shared vulnerabilities, and community-level adaptation strategies. Both methods are designed to generate rich contextual information that complements survey findings. Detailed guides for each instrument (survey, interview, and focus group) are provided in the Supplementary Material (S2–S4 Files).

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Ethical Considerations

This protocol was approved by the Research Ethics Committee of the Federal University of Pará (CAAE: 87492325.6.0000.0018). All participants will provide informed consent

prior to data collection (S1 File – Informed Consent). Identifiable information will be removed during transcription and data processing, and all results will be reported in aggregated form to ensure confidentiality.

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Sample, Inclusion and Exclusion Criteria, and Participant Characteristics

The study will be conducted in the state of Pará, in the Eastern Brazilian Amazon. The selection of this state as the study area is strategically grounded in its centrality to debates on agrarian development, environmental conservation, and climate justice in the Brazilian Amazon³⁶.

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Critically, Pará concentrates approximately 25% of all agrarian reform settlements~~rural~~ settlements in the country, hosting about 27% of all settled families in Brazil, representing the largest absolute population in agrarian reform projects within a single state. This demographic density of settlements, located in the Amazon biome - which is globally recognized as one of the most climate-sensitive - creates a unique and crucial socioecological context for investigation³⁷.

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Focusing on this state allows the examination, on a microscale, of the challenges and adaptation strategies of a vast and vulnerable population, whose experiences are directly shaped by the intersection of land tenure policies, deforestation pressures, and the intensification of climate extremes. Therefore, the results generated from this sample will not only have significant local relevance but will also provide transferable insights to other regions of the Pan-Amazon facing similar dynamics, directly contributing to filling evidence gaps in climate adaptation and mitigation policies grounded in social equity.

To operationalize research in this unique context, the study will be conducted in ten municipalities of Pará, strategically distributed across the three regional superintendencies of the National Institute for Colonization and Agrarian Reform (INCRA).

The choice of 10-15 municipalities is not arbitrary but rather designed to capture the diverse socioecological contexts of agrarian reform settlements in Pará. These municipalities encompass distinct territorial and economic features that are central to understanding adaptation dynamics in the Amazon. They include: (i) areas strongly influenced by mining activities, where extractive industries create both opportunities and vulnerabilities; (ii) regions dominated by agribusiness expansion, with intensive land-use change and pressure on family farming; (iii) floodplain and dryland environments, which present contrasting climate risks and livelihood strategies; (iv) municipalities intersected by major highways such as the Transamazon Highway, where infrastructure development shapes access to markets and exposure to environmental impacts; and (v) extractive and conservation-oriented settlements, where socio-biodiversity chains and traditional livelihoods play a central role.

By sampling across this range of contexts, the study captures key socioecological environments characteristic of Amazonian rural settlements. The selection is not numerically balanced across all contexts; rather, municipalities were purposively chosen to ensure representation of major settlement types and territorial dynamics relevant to climate adaptation. This approach enhances analytical robustness and supports the transferability of findings.~~the study ensures that the results reflect not only average conditions but also the structural heterogeneity of Amazonian rural settlements, thereby~~

enhancing both the analytical robustness and the transferability of findings to other regions facing similar challenges.

In each selected municipality, up to two agrarian reform settlements will be sampled, ensuring representation of the main settlement modalities nationally recognized: Settlement Project (PA), Agroextractive Settlement Project (PAE), Extractive Reserve (RESEX), Sustainable Development Project (PDS), and Collective Settlement Project (PAC). This approach will enable a robust comparative analysis of how different institutional designs and production bases mediate community vulnerability and adaptive capacity in the face of climate change.

Selection of settlements

Eligible settlements must: ~~have~~ (i) have between 20 and 500 settled families, ensuring comparability and logistical feasibility; (ii) reflect production and socioeconomic conditions representative of the municipality; and (iii) be in a stage of creation, installation, structuring, or consolidation ~~(0 to 7, according to INCRA's classification)~~. Settlements with very particular or atypical conditions, such as experimental areas or arrangements that are not replicable, will be excluded.

Participant characteristics

Participants include: (i) local public managers (1–3 per municipality), responsible for municipal policies on agriculture, environment, or rural development; (ii) community leaders (8–12 per settlement), invited to participate in focus groups, representing different social and productive segments; and (iii) settled families (up to 50 per settlement).

covering a diversity of socioeconomic, productive, and demographic profiles. The study expects to include family farmers across different age groups, education levels, family compositions, and modes of engagement in local markets, as well as varying degrees of dependence on natural resources and participation in public policies. In each household, the respondent will be the adult member primarily responsible for agricultural and livelihood decisions (typically the main decision-maker), as this person is expected to have the most comprehensive knowledge of production, income sources, and climate-related impacts.

This sampling strategy seeks to balance territorial representativeness, social diversity, and logistical feasibility, enabling comparisons across municipalities, settlement types, and family profiles.

Selection of families

In each selected settlement, up to 50 families will participate in the study, chosen through simple random sampling from official lists provided by local associations or INCRA. The inclusion criteria for families are: (i) residing in the settlement; and (ii) having an adult household member available to respond to the questionnaire; and (iii) providing voluntary informed consent (ICF) (S1 File – Informed Consent Form). Families will be excluded if: (i) they do not meet the residency requirement; (ii) they refuse or cannot provide consent; or (iii) communication barriers make the interview unfeasible.

~~Participant characteristics~~

~~Participants include: (i) local public managers (1–3 per municipality), responsible for municipal policies on agriculture, environment, or rural development; (ii) community leaders (8–12 per settlement), invited to participate in focus groups, representing different social and productive segments; and (iii) settled families (up to 50 per settlement), covering a diversity of socioeconomic, productive, and demographic profiles. The study expects to include family farmers across different age groups, education levels, family compositions, and modes of engagement in local markets, as well as varying degrees of dependence on natural resources and participation in public policies. This sampling strategy seeks to balance territorial representativeness, social diversity, and logistical feasibility, enabling comparisons across municipalities, settlement types, and family profiles.~~

Qualitative saturation criterion

Qualitative data collection will follow the principle of theoretical saturation³⁸, assessed iteratively by the analysis team during coding. As a guiding strategy, we will consider saturation reached when two or more consecutive interviews or focus groups no longer generate new relevant codes. However, given the high heterogeneity of Amazonian settlements (e.g., floodplains, mining zones, forest-based communities), this criterion will be applied flexibly. Data collection may be extended in specific contexts if new themes continue to emerge or if settlement-specific dynamics require further exploration. To ensure robustness, saturation will be evaluated both within settlement types (~~PA, PAE, RESEX, PDS, PAC~~) and across municipalities and will be triangulated with survey findings and field observations to confirm completeness.

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Sample justification and statistical power

The study was designed to include 15 municipalities, up to 30 settlements, and 600–1,000 families. Power calculations were conducted to determine the minimum sample size needed to achieve at least 80% power at a 5% significance level for detecting modest-to-moderate associations in a multilevel framework. Because key drivers of climate resilience and bioeconomy engagement (e.g., market access, institutional presence, ecological context) operate predominantly at the community level, statistical power is primarily determined by the number of settlements rather than the number of families per settlement.

To reflect this hierarchical structure, simulations incorporated a range of plausible intraclass correlation coefficients (ICCs = 0.05, 0.10, 0.15, 0.30), informed by the literature³⁹ and pilot data. Models were specified to match our primary outcomes: continuous outcomes (e.g., ILDI) were evaluated using linear mixed models (LMM), and ordinal outcomes (e.g., the four-level BEI) using cumulative link mixed models (CLMM). For a sample of 600 families across 30 settlements (average cluster size ≈ 20), $\alpha = 0.05$, and ICC = 0.10, the design provides approximately 80% power to detect a standardized mean difference of 0.35 for continuous outcomes, and to detect an odds ratio of around 2.0 per one-unit increase in an ordinal predictor for BEI under a proportional-odds structure.

These analyses demonstrate that the planned sample is adequately powered to detect effects of practical relevance for our primary hypotheses. If logistical constraints arise during fieldwork, priority will be given to maximizing the number of settlements rather

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than increasing the number of families per settlement, as this yields greater power for
between-cluster comparisons. Full simulation scripts will be made available in the
project's OSF repository to ensure computational transparency and reproducibility.
The study was designed to include 10 municipalities, up to 20 settlements, and 600–
1,000 families. Power calculations were conducted using the simr package (version
4.1.1) in R. Simulations assumed a two-level structure (families nested within
settlements) and were run for a range of intraclass correlation coefficients (ICCs: 0.05,
0.10, 0.15) to reflect plausible clustering effects based on prior rural surveys³⁹ and our
pilot data. The simulations modeled a continuous outcome (e.g., ILDI) with a single
binary predictor at the family level (e.g., participation in a bioeconomy chain). For a
sample of 600 families across 20 settlements (average cluster size of 30), a significance
level of $\alpha = 0.05$, and an ICC of 0.10, the design provides 82% power to detect a
standardized mean difference (Cohen's d) of 0.3. The full simulation code will be
included in the project's OSF repository upon study completion to ensure computational
reproducibility.

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Description of All Processes, Interventions, and Comparisons

Data collection will be carried out sequentially in two integrated phases, as detailed
below. The methodological design was refined following a pilot study conducted in the
municipality of Marabá (PA), with the resulting adjustments described to ensure process
transparency.

Phase 1: Secondary Data and Political-Institutional Context Analysis

461 This phase will focus on building a comprehensive diagnosis of the municipal context. It
462 will involve, first, Climate Risk Assessment, through the compilation and analysis of
463 secondary climate indicators widely recognized in the literature (e.g., temperature,
464 precipitation, extreme events) from sources such as the National Institute of Meteorology
465 (INMET) and the National Institute for Space Research (INPE), to characterize the
466 exposure and sensitivity of the sampled municipalities. Second, Public Policy Mapping,
467 through the identification and documentary analysis of municipal and state-level public
468 policies related to climate change, agriculture, and rural development. This will include
469 the analysis of master plans, laws, and decrees, with attention to both actions explicitly
470 labeled as “climate-related” and sectoral actions with indirect impacts on adaptation and
471 mitigation (e.g., sanitation, civil defense, family farming)²⁷. Third, Public Budget Audit,
472 applying a data mining methodology to municipal transparency portals^{34, 35}, which
473 allows categorization and quantification of public expenditures directly or indirectly
474 related to the climate agenda, thus overcoming the limitation of expenditures formally
475 tagged as “climatic”⁴⁰.

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477 **Phase 2: Primary Data Collection with Key Stakeholders**

478 This phase will capture the perceptions, experiences, and strategies of local actors
479 through three instruments: (i) Household Surveys: administration of a structured
480 questionnaire to a sample of families in each settlement, collecting socioeconomic,
481 productive, climate perception, and adaptation strategy data (S2 File – Household
482 Survey); (ii) Focus Groups with Community Leaders in each settlement type: group
483 discussions to obtain a consolidated view of vulnerabilities, conflicts, and collective

adaptation strategies in the settlements, as well as to validate contextual information (S3 File – Focus Group Guide); (iii) Semi-Structured Interviews with Public Managers: interviews with municipal technicians and secretaries to understand perceptions, implementation challenges, and the articulation of climate policies from the perspective of local government (S4 File – Interview Guide).

Post-Pilot Methodological Adjustments.

Testing the instruments in Marabá (PA) resulted in crucial refinements. For Phase 1, the policy and budget analysis methodology was expanded to systematically capture implicit climate actions, as the pilot revealed that most municipal climate responses are driven by sectoral policies (e.g., agriculture, infrastructure) not explicitly labeled as climate related. For Phase 2, the household survey was adapted to include specific modules for riverine and floodplain communities, capturing ~~particular climate~~ climate impacts (e.g., changes in flood cycles, river-based access). The interview guide with managers was revised to replace technical jargon with more accessible terms, reflecting the fact that many municipalities lack a consolidated formal framework for climate policy.

Comparative Analysis Strategy

The analysis will be structured across four comparative levels, allowing the isolation of factors associated with resilience at different scales: (i) Across municipalities, contrasting indicators of vulnerability, climate budget, and policy robustness; (ii) Across settlement types (~~PA, PAE, RESEX, PDS, PAC~~), examining how different land tenure and institutional arrangements mediate convergences ~~ess-of~~ families within settlements,

correlating socioeconomic profiles (education, income diversification) with adoption of adaptation practices; (iv) Across social actors, triangulating the perspectives of public managers and communities to identify convergences and divergences in risk perceptions and policy effectiveness.

This multilevel approach is essential for identifying robust spatial, socioeconomic, and institutional patterns associated with climate resilience.

Data Analysis Plan

Data analysis will follow a sequential explanatory mixed-methods strategy (QUAN → QUAL)⁴¹, in which the initial quantitative analysis will provide an overall picture to be subsequently deepened and contextualized by qualitative investigation. Triangulation of data from different sources (quantitative, qualitative, and secondary) will be employed to validate and enrich the findings, ensuring a robust understanding of the factors that influence climate resilience⁴². All analyses will be performed using R software (version 4.3.0 or higher) for the quantitative component and NVivo software (version 14 or higher) to support the organization and coding of qualitative data.

General Strategy and Integration

The analysis will occur in three main stages, corresponding to the data sets, culminating in an integration phase: (i) Secondary Data and Municipal Context Analysis, to characterize the macro context. (ii) Quantitative Primary Data Analysis (Survey), aimed at identifying patterns and testing hypotheses at household and community levels. (iii) Qualitative Primary Data Analysis (Interviews and Focus Groups), to explore causal

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mechanisms, perceptions, and nuances. (iv) Finally, integration and triangulation, with a synthesis of the findings to answer the research objectives.

Secondary Data and Municipal Context Analysis

The analysis of secondary data will proceed in three integrated steps to construct key municipal-level variables. First, a Municipal Climate Vulnerability Index (MCVI) will be created by compiling secondary indicators of exposure, sensitivity, and adaptive capacity from sources including AdaptaBrasil, MCTI, IBGE, and INPE. These indicators will be harmonized by converting to per capita measures where appropriate, standardized (z-scores), and aggregated through Principal Component Analysis (PCA)⁴³. Second, municipal climate change plans will be assessed using categorical content analysis to generate a plan quality score. Third, a public expenditure audit, employing text mining of municipal transparency portals, will be conducted to identify and quantify climate-related budgets.

To ensure robustness, a multi-step strategy will address data quality challenges. For the MCVI, discrepancies or missing values will be handled using imputation techniques, with sensitivity analyses conducted to assess the impact on the index. Furthermore, all secondary data - including the budget analysis and plan documentation - will be cross-checked against municipal transparency portals and triangulated with qualitative data from interviews and focus groups to ensure contextual accuracy.

Finally, the plan quality scores, and budget allocations will be combined to create a Municipal Climate Engagement Typology (e.g., High, Medium, Low). Both the MCVI

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and this engagement typology will be incorporated as contextual variables in the multilevel models.

Quantitative Primary Data Analysis (Survey)

Initially, descriptive statistics (frequencies, means, medians, and standard deviations) will be used to characterize the sample profile. Subsequently, to operate key concepts, composite indices will be constructed, and their internal consistency will be assessed using Cronbach's alpha ($\alpha > 0.7$) and/or Confirmatory Factor Analysis (CFA)⁴⁴. The planned indices include Income and Livelihood Diversification Index (ILDI); Climate Adaptation Practices Index (CAPI); Bioeconomy Engagement Index (BEI); and Climate Vulnerability Perception Scale (CVPS).

To test the hypotheses in Table 1, linear mixed-effects models (LMM) and generalized linear mixed-effects models (GLMM)⁴⁵ will be applied, which are appropriate for hierarchical data (families nested within settlements and municipalities). The MCVI and the municipal engagement typology will be included as higher-level variables (level 2), while family characteristics (level 1) will be used as predictors and control variables.

Addressing Confounding and Sensitivity Analysis

To mitigate potential confounding, the multilevel models will include a rich set of pre-specified covariates at different levels: (1) Household Level (Level 1): Age and education of the household head, household size, gender of the household head, asset index, length of residence in the settlement; (2) Settlement Level (Level 2): Settlement type, distance to the nearest paved road, presence of a producers' cooperative, years since settlement

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establishment; (3) Municipality Level (Level 3): Municipal Climate Vulnerability Index (MCVI), Municipal Climate Engagement Typology, and geographic region (e.g., floodplain vs. upland). Acknowledging the impossibility of measuring all potential confounders, we will conduct a formal sensitivity analysis for the key associations related to our primary hypotheses (e.g., the association between BEI and household income). We will use the `sensemakr` package in R, following the framework developed by Cinelli and Hazlett (2025)⁴⁶. For each key finding, we will report the Robustness Value (RV), the minimum strength of association (partial R^2) that an unmeasured confounder would need to have with both the exposure and the outcome to explain away the observed effect. We will also present contour plots to visualize how the effect estimate and its confidence interval would change under different scenarios of confounding. This will provide readers with a transparent assessment of the robustness of our inferences to unobserved confounding.

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Operationalization of the Indices

To ensure transparency and replicability, each index will be constructed according to clear theoretical and statistical criteria, as follows:

- Income and Livelihood Diversification Index (ILDI): formative index calculated using the Herfindahl-Hirschman formula ($1 - \sum p_i^2$)³¹ based on the relative shares of agricultural, extractive, and non-agricultural income. It will not be evaluated by Cronbach's alpha but described statistically.
- Climate Adaptation Practices Index (CAPI): reflective index, constructed from the standardized sum of reported adaptation practices (e.g., irrigation, resistant

varieties, collective work). Internal consistency will be assessed by Cronbach's alpha³² ($\alpha \geq 0.7$) and, when appropriate, by confirmatory factor analysis.

- Bioeconomy Engagement Index (BEI): formative ordinal variable, with increasing levels of engagement in bioeconomy value chains (0: no participation; 1: primary production; 2: processing; 3: processing + certification/market differentiation). Higher values indicate greater household-level involvement. For multilevel analyses, the BEI will be used primarily as a level-1 predictor (household-level) to examine associations with income, adaptive capacity, and adoption of mitigation/adaptation practices. To explore community-level patterns, the BEI will also be summarized at the settlement level using the mean or distribution of household scores (e.g., proportion of households at each engagement level). These aggregated measures will be used as level-2 contextual variables when relevant for hypotheses concerning settlement-level dynamics (e.g., HP5).
- Climate Vulnerability Perception Scale (CVPS): reflective scale, constructed as the average of Likert-type items (1–5) capturing perceptions of climate risk (drought, flood, crop loss, food insecurity). Internal consistency will be assessed by Cronbach's alpha ($\alpha \geq 0.7$) and unidimensionality by CFA.
- Municipal Climate Vulnerability Index (MCVI): formative index based on indicators of exposure (climate), sensitivity (e.g., agricultural dependence, poverty), and adaptive capacity (e.g., education, policy access). It will be constructed from standardized indicators (z-scores) and aggregated through PCA.

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620 PCA adequacy indicators ($KMO > 0.6$) and variance explained by the first
621 component will be reported.

622 Each index will be statistically validated, described in terms of distribution, central
623 tendency, and dispersion, and interpreted according to its reflective or formative nature.

624

625 **Qualitative Primary Data Analysis**

626 Transcriptions of interviews and focus groups will be analyzed using Braun and Clarke's
627 Reflexive Thematic Analysis⁴⁷ in six stages: (1) familiarization; (2) generation of initial
628 codes; (3) search for themes; (4) review of themes; (5) definition and naming of themes;
629 and (6) report production. This analysis will identify central themes related to local
630 perceptions, barriers, facilitators, and understandings of climate change, adaptation, and
631 bioeconomy.

632 All qualitative findings will be reported according to the COREQ guideline (or SRQR,
633 where applicable), ensuring comprehensive documentation of data collection and analytic
634 procedures.

635

636 **Integration and Triangulation**

637 In the final phase, the findings will be integrated following a sequential explanatory
638 mixed-methods design (QUAN → QUAL) and a joint display triangulation
639 framework through a triangulation matrix⁴². The integration will be reported in
640 accordance with the Good Reporting of a Mixed Methods Study (GRAMMS) framework
641 and the Standards for Reporting Qualitative Research (SRQR) guidelines for the
642 qualitative component.

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643 ~~This~~ A triangulation matrix will be constructed to systematically map convergences,
644 complementarities, and divergences between quantitative, qualitative, and secondary data
645 findings for each research objective and hypothesis~~and secondary results for each~~
646 ~~research objective and hypothesis~~. The matrix will define decision rules for interpreting
647 relationships: (I) Convergence: Findings from different methods provide corroborating
648 evidence for the same conclusion; (II) Complementarity: Findings from different methods
649 elucidate distinct, non-overlapping aspects of a phenomenon, together providing a richer,
650 more complete picture; (III) Discordance: Findings from different methods are
651 contradictory or inconsistent. In such cases, we will re-examine the data and the
652 assumptions of each method to propose potential explanations for the discrepancy (e.g.,
653 differences in perspective between managers and farmers, social desirability bias in
654 surveys). (IV) Silence: An expected finding from one method is absent in the data from
655 another method

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656 For example, a quantitative pattern of low adoption of a bioeconomy practice (QUAN)
657 will be examined considering reported barriers in interviews (QUAL) and the
658 (non)existence of supportive policies identified in secondary data. This synthesis will
659 enable not only the identification of *what* occurs but also *why* and *how* adaptation and
660 resilience processes unfold.

661

662 **Handling of Missing Data**

663 The strategy for handling missing data will be determined by its mechanism and extent.
664 During data collection, digital forms with mandatory field checks will minimize
665 inadvertent missingness. Interviewers will be trained to probe for reluctant or "don't

know" responses. Prior to analysis, we will generate a missingness pattern report. Little's
 Missing Completely at Random (MCAR)⁴⁸ test will be used to inform the choice of
 handling method, though we will primarily assume data is Missing at Random (MAR).
 Our primary analysis will use a Full Information Maximum Likelihood
 (FIML) estimation within the mixed-effects models (lme4 package in R). FIML uses all
 available data points without discarding partial cases and provides unbiased estimates
 under the Missing at Random (MAR) assumption.
 To assess the robustness ~~of our inferences to the MAR assumption~~, we will conduct
 sensitivity analyses using ~~multiple~~ Multiple Imputation (mice package in R). We will
 create 20 imputed datasets, including all model variables and auxiliary variables
 correlated with missingness. The multilevel structure will be respected during imputation
 using the ~~2lonly~~ method, incorporating cluster means of level 1 variables as predictors.
All model variables, plus auxiliary variables correlated with missingness (e.g., interview
duration, interviewer ID), will be included in the imputation model. Results from the
 pooled imputed models will be compared with the primary FIML analysis.
~~The amount and pattern of missing data for all key variables will be reported in the final~~
~~publications.~~ In addition to primary data, we will assess and report missingness in all
secondary datasets, including climate indicators, socioeconomic data, municipal plans,
and budget information. For missing values in external data sources, we will apply
appropriate strategies depending on the mechanism: (i) triangulation with alternative
official databases when equivalent indicators exist, (ii) imputation of continuous
variables using multilevel or time-series-based methods when justified, and (iii) explicit
coding of structural missingness (e.g., municipalities with no climate plans or no

identifiable climate-related expenditures). The extent and implications of missing secondary data will be documented in the Municipal Climate Vulnerability Index (MCVI) and in the construction of the Municipal Climate Engagement Typology (MCET), and sensitivity checks will be conducted to assess the robustness of results to these gaps

Data Management Plan

~~A comprehensive data management plan has been developed to ensure the integrity, reproducibility, and controlled access of the data, while guaranteeing strict compliance with ethical protocols. All secondary data from public domains will be meticulously documented and cited. Primary qualitative data (audio recordings and transcripts) and survey data, due to their sensitive nature and the risk of re-identification in communities with extensive sampling, will undergo a rigorous advanced anonymization process. This process includes the generalization of indirect identifiers and the suppression of excessively specific details.~~

~~Anonymized survey data, along with processing and analysis scripts and full documentation (metadata, data dictionary), will be deposited in the Open Science Framework (OSF) repository under a 12-month embargo from the date of publication of the primary results. After this period, the data will be made openly available. During the embargo, access requests can be submitted via the OSF platform or by emailing the corresponding author. Requests will be reviewed by the steering committee and granted upon approval of a brief research proposal and the signing of a data use agreement that prohibits re-identification attempts.~~

Raw qualitative data (audio and identifiable transcripts) will remain permanently confidential and restricted exclusively to the research team, as approved by the Ethics Committee. This framework seeks to balance the principles of scientific transparency and reproducibility with the fundamental protection of research participants' privacy and confidentiality.

All secondary data comes from public sources and will be fully documented. Primary surveys and qualitative data will undergo rigorous anonymization to protect participant confidentiality. Anonymized datasets, metadata, and analysis scripts will be deposited in the Open Science Framework (OSF) under a temporary embargo; raw qualitative data (audio or identifiable transcripts) will remain confidential as required by ethics guidelines. Detailed documentation will accompany all shared materials to ensure transparency and reproducibility.

Ethical Considerations

This study protocol was approved by the Research Ethics Committee (CEP) of the Federal University of Pará (UFPA), in accordance with Resolution No. 510/2016 of the National Health Council, under CAAE number 87492325.6.0000.0018 (S5 File—REC Approval).

The Informed Consent Form (ICF) will be obtained from all participants prior to their inclusion in the study (S1 File—Informed Consent Form). For community members, the ICF will be provided in writing, reading aloud, and explained in clear and accessible language, considering possible variations in literacy levels. For public managers, the ICF will be presented in written form. All participants will be informed about the research

~~objectives, the procedures involved, potential risks and benefits, the voluntary nature of participation, and their right to withdraw consent at any time without penalty. Written consent will be obtained by signature or fingerprint.~~

~~To ensure confidentiality, all data collected will be anonymized. Personal identifiers (names, specific locations, positions) will be removed and replaced with codes during transcription and data analysis. The digital database will be stored on a password-protected computer, and physical documents will be kept in a locked cabinet with restricted access to the research team. Results will be presented in aggregate form, ensuring that no individual participant can be identified.~~

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Study Status and Timeline

~~The project is structured into three main phases, distributed over 36 months (2025–2027). As of the submission date of this protocol, the status of the study is as follows: ethical approval has been obtained, and a pilot study was conducted in the municipality of Marabá (PA) exclusively to refine the data collection instruments. Participant recruitment for this pilot occurred from September 08 to 26, 2025, and all participants provided written informed consent. Large-scale data collection, as detailed in this protocol, has not yet started and is scheduled to run from November 03, 2025 to December 20, 2026.~~

~~One of the key factors contributing to the success of the pilot study in Marabá (PA) was the strong support and trust of local communities. This was made possible because several members of the research team were themselves selected from these communities, ensuring cultural familiarity, credibility, and direct involvement in fieldwork. In addition,~~

the study has established formal and informal partnerships with local organizations, including universities, social movements, and civil society associations, which provide logistical support, facilitate dialogue with community leaders, and strengthen the ethical dimension of the research. These arrangements are not limited to Marabá (PA); rather, they extend to the other municipalities included in the study, where team members already have long-standing connections. This embedded approach enhances both the feasibility of data collection and the legitimacy of the research process, while reinforcing the principle of co-production of knowledge with local stakeholders.

The detailed timeline of activities is presented in Table 2 below.

Table 2—Development and Implementation of the Study Protocol (2025–2026)

Etapas	Year 1				Year 2				Year 3			
	1T	2T	3T	4T	1T	2T	3T	4T	1T	2T	3T	4T
Preparation	X	X	X	X								
Instruments	X	X										
Recruitment			X	X								
Training			X	X								
Data Collection (Pilot)			X									
Adjustment			X									
Field Research				X	X	X	X	X	X			
Formation of Local Teams				X	X							
Data Collection				X	X	X	X	X				
Data	X	X	X	X	X	X	X	X	X			
Secondary Data Survey	X	X										
Primary Data Curation			X	X	X	X	X	X	X			
Data Analysis	X	X	X	X	X	X	X	X	X			

Dissemination											X	X	X
Technical Reports											X		
Academic Articles											X		
Workshops											X	X	
Policy Brief											X		X

To ensure feasibility across a large and heterogeneous territory, the project relies on research teams established in each of the 10 selected municipalities. These teams are composed of local researchers and students, many of whom are members of the communities themselves, which facilitates trust and logistical access. The study is fully supported by funding from CNPq (Call 19/2024 — Pro Amazônia), which provides resources for fieldwork, data collection, and training. In addition, the protocol is strengthened by formal partnerships with local universities, social movements, and civil society organizations, which contribute infrastructure, mobilization, and institutional legitimacy. This decentralized structure, combined with strong local support, substantially reduces the logistical risks of conducting fieldwork in remote Amazonian areas and ensures the viability of the proposed design.

The preparation stage involves finalizing the data collection instruments based on insights gained from the pilot study, training the research team in ethical and methodological procedures, and establishing partnerships and logistical arrangements to ensure access to the study sites. Field research entails the systematic collection of secondary data, including municipal indicators, documentary evidence, and budget analyses, conducted in parallel with the collection of primary field data through household surveys, focus groups, and semi-structured interviews in the ten selected

municipalities. Data analysis includes the curation, organization, and cleaning of datasets, followed by the statistical analysis of quantitative data, thematic analysis of qualitative data, and the integration and triangulation of findings using a mixed methods framework. Dissemination comprises the preparation of manuscripts for submission to peer-reviewed scientific journals, the development of technical reports and policy briefs tailored for public managers, and the production of specific materials designed for local policymakers and community stakeholders. In addition, workshops will be organized to return and discuss the study's findings with the participating communities, ensuring that the results inform both academic debates and practical decision-making at the local level. A pilot study was conducted in Marabá (PA) to refine instruments and procedures. Large-scale data collection is scheduled for December 2025 to December 2026, followed by data analysis and dissemination activities in 2027. These timelines reflect logistical constraints of fieldwork in remote Amazonian settlements.

Discussion

Study Limitations

The main methodological advancement of this protocol lies in the integration of climate resilience indicators with bioeconomy dynamics in agrarian reform settlements. To the best of our knowledge, this approach has not been systematically applied in international studies, which underscores the novelty and global relevance of this work. However, some limitations are anticipated. First, the reliance on secondary data is subject to the inherent limitations of public databases, such as variations in quality, completeness, and frequency

of updates across municipalities. We will seek to mitigate this limitation through triangulation of data sources whenever possible.

Second, the qualitative component, based on interviews, may be subject to social desirability bias, in which participants provide responses they believe are expected by researchers rather than their genuine perceptions. To minimize this risk, interviewers will be trained to build rapport and foster a comfortable environment, assuring participants that there are no right or wrong answers.

Finally, the intentional selection of municipalities and settlements may limit the generalizability (external validity) of the results to the entire state of Pará. The study aims to provide an in-depth analysis of specific contexts, rather than a statistically representative overview. Nevertheless, the detailed methodological description will allow an assessment of the transferability of the findings to similar contexts.

The main methodological advancement of this protocol lies in the integration of climate resilience indicators with bioeconomy dynamics in agrarian reform settlements. To our knowledge, this approach has not been systematically applied in international studies, reflecting both the novelty and global relevance of this work. Nevertheless, several limitations should be acknowledged, together with justifications for the chosen design.

First, regarding study design, the mixed-methods and multilevel analytical approach was selected because climate adaptation and bioeconomy practices are shaped simultaneously by household characteristics, community organization, and municipal institutional capacity. A purely quantitative design would not capture the mechanisms, narratives, and context-specific meanings that influence adaptive behavior, whereas a purely qualitative design would not allow systematic comparison across municipalities and settlements. The

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832 sequential explanatory design (QUAN → QUAL), supported by triangulation, therefore
833 provides an appropriate structure for addressing the research questions. Alternative
834 designs were considered, particularly a longitudinal panel design and an experimental or
835 quasi-experimental evaluation of policy interventions, but were not feasible due to the
836 absence of comparable baseline data, the difficulty of randomization in this context, and
837 the logistical challenges of multi-year fieldwork in remote Amazonian territories.
838 Second, the reliance on secondary data introduces constraints related to data availability,
839 variable quality, and inconsistent reporting across municipalities. Climate indicators,
840 budget classifications, and policy documents vary in completeness and temporal
841 coverage. To mitigate these limitations, the study incorporates (i) triangulation across
842 multiple official databases, (ii) imputation and sensitivity analyses for incomplete
843 indicators, and (iii) explicit treatment of structural missingness (e.g., municipalities with
844 no climate plans or no identifiable climate-related expenditures).
845 Third, the qualitative component may be vulnerable to social desirability and
846 positionality biases. Interviewers may provide responses they perceive as desirable or
847 aligned with institutional expectations. This risk will be minimized through careful
848 interviewer training, rapport-building, and emphasis on confidentiality and the absence of
849 “right” or “wrong” answers. Additionally, by drawing on community-based researchers
850 familiar with the local context, the study enhances trust and reduces outsider bias.
851 Fourth, the intentional selection of municipalities and settlements limits the
852 generalizability of findings for the entire state of Pará. The study does not aim to produce
853 a statistically representative estimate of all settlements; instead, it provides an in-depth
854 analysis of key socioecological contexts. However, the detailed methodological

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855 description and the diversity of selected municipalities allow assessment of the
856 transferability of insights to similar contexts across the Eastern Amazon.
857 Fifth, although the protocol draws on international literature on climate adaptation,
858 resilience, and rural livelihoods, it extends this body of work in important ways. Previous
859 studies often examine adaptation or bioeconomy pathways separately; few integrate both
860 dimensions with a multilevel analytical structure and grounded qualitative evidence. The
861 protocol also adapts global frameworks (e.g., livelihood diversification, adaptation
862 portfolios, municipal climate governance) to the institutional realities of agrarian reform
863 settlements, an understudied but globally relevant category of rural territories.
864 Finally, the potential applications of the findings justify the chosen approach. Once
865 completed, the study will produce evidence directly applicable to municipal and state
866 policymakers in Brazil, supporting the refinement of climate adaptation and bioeconomy
867 strategies, the allocation of public resources, and the design of territorial programs.
868 Internationally, the protocol can inform studies in other regions with decentralized
869 climate governance, rural poverty, and socio-biodiversity economies, such as Southeast
870 Asia, Central America, and parts of Sub-Saharan Africa. With minor contextual
871 adaptations (e.g., indicator selection, policy frameworks, and institutional architecture),
872 the methodological approach can be generalized to rural areas undergoing similar
873 socioecological transitions.
874 Overall, while the study faces limitations inherent to its observational and context-
875 specific design, the mixed-methods, multilevel structure, combined with transparency,
876 triangulation, and sensitivity analyses, provides a robust framework for advancing
877 empirical knowledge on climate resilience and bioeconomy pathways in vulnerable rural

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regions. In the final analysis, the implications of each limitation will be explicitly assessed through sensitivity analyses, triangulation, and transparent reporting of uncertainty, and these will be clearly communicated in the presentation of results to ensure that conclusions accurately reflect the strength and boundaries of the evidence.

Dissemination Plan

The results of this research will be disseminated through multiple channels to reach academic audiences, public policy managers, and the participating communities themselves.

For the academic audience, the primary results will be submitted for publication in international open access, peer reviewed journals (such as *PLOS Climate*, *Climate Policy*, or *Sustainability*) as well as in national journals. The findings will also be presented at national and international scientific conferences in the fields of climate change, public policy, and Amazonian studies.

Dissemination to public managers and policymakers is a central point of this research and will be carried out through the preparation, in Portuguese, of a policy brief (executive summary) and a technical summary report in non technical language specific to each municipality. These documents will be distributed to key actors, including the Pará State Secretariat for the Environment and Sustainability (SEMAS-PA), the State Secretariat of Agriculture (SAGRI-PA), and the municipal governments involved in the study. The goal is to directly inform the design and implementation of more effective climate policies.

Finally, in the participating communities, the results will be returned to the settlements through community workshops and the production of educational booklets. These

workshops will be conducted in an accessible format, fostering dialogue about the results and discussing potential community adaptation strategies. This action is crucial to honor participants' contributions and ensure that the research generates local benefits.

Protocol Amendments and Early Termination

Any substantial modification to the study protocol—including changes in methodology, sample size, or participant eligibility criteria—will be submitted for approval by the Research Ethics Committee (CEP) prior to implementation. All changes will be documented in a master project file.

The study may be interrupted before its planned completion for the following reasons: (a) identification of unforeseen serious risks to participants or researchers; (b) insurmountable practical difficulties in data collection (e.g., inability to access key communities or obtain public data); (c) formal withdrawal of consent by partner communities or local authorities.

If fieldwork constraints arise, we will reduce the sample to 6 municipalities while ensuring representation of settlement types.

In the event of early termination, the Principal Investigator will formally notify the Ethics Committee and the funding agency (CNPq). All data collected up to the point of termination will be anonymized, securely stored, and may be used for analysis, with the circumstances of the interruption clearly disclosed in any resulting publication.

Author's Contributions

Conceptualization: Daniel Silva.

924 Data curation: Vitor Castro.
925 Formal analysis: Larissa Alves, Emilio Mendes.
926 Funding acquisition: Daniel Silva.
927 Investigation: Larissa Alves, Naurinete Reis, Maclem Erane, Emílio Mendes.
928 Methodology: Daniel Silva, Larissa Alves, Naurinete Reis, Maclem Erane, Emilio
929 Mendes.
930 Project administration: Daniel Silva.
931 Software: Vitor Castro
932 Supervision: Daniel Silva.
933 Validation: Naurinete Reis.
934 Visualization: Emilio Mendes.
935 Writing – original draft: Daniel Silva.
936 Writing – review and editing: Daniel Silva, Larissa Alves, Naurinete Reis, Maclem
937 Erane, Emilio Mendes, Vitor Castro.
938

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942 fundamental to the data collection for this study. We also thank the National Council for
943 Scientific and Technological Development (CNPq) for the financial support that made
944 this research possible ~~through Call 19/2024 Pro-Amazônia~~.
945

946 **Supporting Information**

947 ~~S1 File. Methodology for the analysis of budget data~~
 948 ~~S2-S1~~ File. Focal Groups Guide
 949 ~~S3-S2~~ File. Interview Guide for Public Managers
 950 ~~S4-S3~~ File. Questionnaire Guide for Family Farmers
 951 ~~S5 File. Informed Consent Form (ICF) used~~
 952 ~~S6 File. Research Ethics Committee (REC) approval — the official document certifying~~
 953 ~~project approval~~
 954

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1090

REVIEWER #1

We would like to thank you for the careful reading of our manuscript and for the constructive and insightful comments. We appreciate the positive evaluation regarding the clarity, relevance, and overall structure of the study. Following your guidance, we have revised the manuscript extensively. The changes have substantially improved methodological transparency, theoretical grounding, clarity of objectives, and the international relevance of the protocol.

Below we provide a detailed, point-by-point response to all comments. All revisions have been incorporated in the updated manuscript.

1. Need for more citations in the Methods section

Reviewer comment:

“There is a lack of citations along the methods section... especially regarding the calculation of indexes and approaches for missing data analysis...”

Response:

We agree and have added multiple methodological references throughout the Methods section, including citations related to diversification measures, reflective and formative index construction, PCA, mixed-effects modeling, multiple imputation, and multilevel missing data handling. These additions strengthen transparency and replicability.

2. Need to summarize / merge certain sections (Data management, Ethics, Timeline)

Reviewer comment:

“Some sections in the methods could be significantly summarised and merged... data management, ethical considerations, project timeline.”

Response:

We have condensed these sections substantially:

- Ethical Considerations was moved to the Methods section and shortened.
- Data Management was reduced to a concise paragraph describing anonymization, storage, and OSF deposition.
- Study Status and Timeline was reduced and summarized in a brief paragraph.

This improves the manuscript's focus on scientific content.

3. Discussion needs expansion (design rationale, contributions, international relevance)

Reviewer comment:

“The discussion needs to be significantly expanded... explain why the chosen study design is appropriate... highlight how the protocol builds upon previous studies... describe international relevance...”

Response:

The Discussion has been substantially expanded to include:

- A clear justification for the mixed-methods, multilevel design.
- An explanation of alternative designs (e.g., panel, quasi-experimental) and why they were not feasible.
- A comparison with existing literature showing how our protocol integrates climate adaptation, bioeconomy, governance, and resilience.
- A new section on the international relevance of the study and pathways for generalization.
- A description of how limitations will be accessed and communicated in the final results.

Sections that were not directly scientific (e.g., dissemination plan, early termination) were removed.

4. Need for a subsection describing qualitative data collection**Reviewer comment:**

“There should be a subsection prior to ‘Sample inclusion...’ that describes them.”

Response:

We added a new subsection titled “Data Collection Methods” before the sampling description, clarifying procedures for:

- household surveys
- semi-structured interviews
- focus groups

This ensures coherence with the qualitative saturation criteria.

5. Clarification of agrarian reform settlements and ‘settled families’**Reviewer comment:**

The reviewer did not understand these terms.

Response:

We added a clear definition in the Introduction and Sample section explaining that agrarian reform settlements are rural territories created through Brazil's federal land redistribution program (INCRA), and "settled families" refers to beneficiary households receiving land through this program. This addresses ambiguity for international readers.

6. Clarifications on indicator descriptions (range, direction, origin, aggregation)**Reviewer comment:**

The reviewer requested more details for ILDI, CVPS, BEI, RACI, MCET.

Response:

We revised all indicators to include:

- range of values
- expected direction (associational)
- conceptual grounding
- whether original or adapted
- aggregation from household to settlement level (when applicable)

Descriptions are now consistent and transparent.

7. Avoiding overstatements (e.g., "no standardized methodology")**Reviewer comment:**

The initial phrasing overstated the absence of methods.

Response:

We rewrote this section to present a more nuanced description, noting that existing methods are fragmented and unevenly applied, and clarifying that our protocol contributes by integrating climate adaptation, resilience, and bioeconomy in a multilevel framework.

Conclusion

We thank the reviewer and Academic Editor once again for the constructive feedback that greatly improved the clarity and contribution of the protocol. We respectfully resubmit the revised manuscript for further consideration.

REVIEWER #2

We thank you sincerely for your detailed and insightful review. Your comments significantly improved the methodological clarity, rigor, and overall contribution of our study protocol. We have carefully addressed every point raised and revised the manuscript accordingly. Below we provide a point-by-point response, following the structure of your comments.

1. Clarification of causal vs. associational language

Reviewer comment:

“The current framing occasionally suggests causal claims... Unless you add a credible identification strategy, the objectives and hypotheses should be clearly associational.”

Response:

We agree and have revised the entire manuscript to remove any causal language. Specifically:

- All hypotheses (PH1–PH5) were rewritten using associational terms (“associated with,” “co-occur with,” “positively/negatively correlated with”).
- We removed verbs implying causality (“leads to,” “results in,” “increases,” “reflects”), replacing them with neutral correlational phrasing.
- The Objectives section was revised to explicitly state that the study aims to identify associations, not causal effects.
- We clarified the observational nature of the design in the Study Limitations.

These changes ensure full alignment with observational study and PLOS ONE guidance.

2. Pre-specification of confounders across levels & sensitivity analysis (Cinelli & Hazlett)

Reviewer comment:

“Pre-specify rich confounders... adopt Carlos Cinelli’s framework for sensitivity analysis... robustness values and contour diagnostics.”

Response:

We thank the reviewer for this excellent suggestion. We incorporated the following revisions:

- Added a new subsection explicitly listing confounders at the household, settlement, and municipal levels (market access, infrastructure, ecological

context, cooperatives, local associations, institutional capacity, political economy, and geographic factors).

- Added a detailed Sensitivity Analysis Plan, adopting the Cinelli & Hazlett ¹ framework, with robustness values and contour plots to quantify the impact of unobserved confounding.
- Specified that analyses will be conducted using sensemakr, and results will report E-values/robustness thresholds.
- Clarified that this sensitivity analysis complements multilevel models and multiple imputation procedures.

These additions substantially strengthen the transparency and rigor of the analytical plan.

3. Expanded power calculations for multilevel design

Reviewer comment:

“Power deserves expansion to reflect clustering... ICC around 0.10... variation between communities... ordinal BEI harder to power... redo power and MDE calculations... grid of plausible ICCs... consider number of communities vs. number of respondents.”

Response:

We fully revised the Power Calculations section to incorporate:

- A clear explanation of the role of ICCs and design effects.
- Simulations for continuous, binary, ordinal (CLMM), and count outcomes across ICCs = 0.05, 0.10, 0.20, 0.30.
- A statement that power is driven primarily by the number of settlements, not households.
- Justification for including 20–30 settlements in the sampling design.
- Minimum detectable effect estimates for:
 - LMM (continuous),
 - CLMM (BEI ordinal predictor/outcome),
 - and logistic models (binary outcomes).
- Discussion of trade-offs (increasing clusters > increasing respondents).

We revised the introductory sentence to emphasize that power calculations were used to determine the minimum sample size required, not merely to illustrate tool usage.

4. Explicit clarification of where variation is expected to come from

Reviewer comment:

“It would help to be explicit about where you expect variation to come from... most identifying variation will be between communities.”

Response:

We added explicit statements in both the Sampling Strategy and Quantitative Analysis sections that:

- Most identifying variation is expected at the settlement (level-2) scale, given the importance of market integration, cooperative presence, mobility, and ecological context.
- This expectation guided our choice to maximize the number of settlements and informed the multilevel modeling approach.

This cross-referencing improves internal coherence.

5. Clarification of ordinal modeling for BEI

Reviewer comment:

“The move from a binary to a four-category engagement index is sensible... commit to an analysis approach appropriate for ordered categorical.”

Response:

We revised the Analytical Plan to state explicitly:

- BEI will be modeled using cumulative link mixed models (CLMM) when used as an outcome.
- BEI may serve as a level-1 or level-2 predictor, with appropriate aggregation (means or distributions) at the settlement level when required.
- The proportional odds assumption will be checked.

This ensures appropriate treatment of the indicator.

6. Missing data plan, with clarity across levels

Reviewer comment:

“Please indicate how you will handle missingness at each level... multiple imputation respecting clustering... reweighting strategies... sensitivity.”

Response:

We expanded the Missing Data section by:

- Differentiating missingness across household, settlement, and municipal levels.
- Specifying multilevel multiple imputation using mice and pan, with clustering preserved.
- Including inverse probability weighting (IPW) and augmented IPW for differential nonresponse.
- Adding procedures for secondary data missingness (climate indicators, municipal plans, budgets).
- Clarifying that missingness in municipal plans/budgets may be structural and analytically meaningful.
- Integrating this with the Cinelli-style sensitivity analysis.

This strengthens the rigor of the protocol.

7. Reduction of researcher degrees of freedom and primary outcomes

Reviewer comment:

“Pre-specifying a short list of primary outcomes and hypotheses... identify explicitly which variables will be used.”

Response:

We created a new paragraph at the beginning of Primary Outcomes:

- Specifying exactly four primary outcomes (ILDI, CVPS, BEI, RACI).
- Listing which hypotheses (PH1–PH5) each primary outcome relates to.
- Clearly separating expected direction (conceptual meaning) from hypothesis testing (associational relationships).

Secondary outcomes are now explicitly labeled and described as complementary.

8. Qualitative standards (COREQ/SRQR) and mixed-methods integration (GRAMMS)

Reviewer comment:

“Commit to COREQ/SRQR... provide a GRAMMS-style description... triangulation table.”

Response:

We added:

- A clear statement that qualitative reporting will follow COREQ.
- A full GRAMMS-consistent description of integration procedures.
- A triangulation table with decision rules for convergence, complementarity, and discordance.
- Updates to the Saturation and Qualitative Methods sections.

These revisions significantly clarify the role of qualitative data.

Conclusion

We sincerely appreciate the reviewer's contributions. The manuscript has improved substantially because of this detailed and thoughtful review. We believe the revised version now aligns fully with PLOS ONE's expectations for study protocols and provides a stronger, clearer, and more rigorous analytical framework.

We hope the revised manuscript meets with your approval, and we thank you again for your constructive feedback.

Sincerely,

Corresponding Author